

Conformal universality of bulk Ising correlations under weak interactions

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Abstract

We prove conformal universality of mixed bulk spin and energy correlations for the square-lattice Ising model with sufficiently small, square-symmetric, finite-range even multispin perturbations of either sign. One critical-temperature branch and two field normalizations apply in every bounded simply connected C^2 Jordan domain with free, plus, or minus boundary conditions, and in the periodic thermodynamic plane state. Energies are centered in their actual states.

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1 Introduction and main result

We establish mixed bulk spin and energy universality for small signed finite-range perturbations of the critical square-lattice Ising model. We first specify the model and the common normalization in the theorem.

Fix $J > 0$. A potential assigns a real number $V(A)$ to each nonempty finite set $A \subset \mathbb{Z}^2$. We assume that $V(A) = 0$ unless $|A|$ is even and $\text{diam } A \leq R_V$ for some fixed finite R_V , and that V is invariant under every lattice translation, reflection, and rotation by $\pi/2$. There are only finitely many shapes modulo translation. No positivity assumption is imposed on their coefficients. Throughout the proof V is fixed.

Identify $\mathbb{R}^2$ with $\mathbb{C}$. Let D be a bounded simply connected domain with C^2 Jordan boundary, and put $D_a = D \cap a\mathbb{Z}^2$. The spin configuration is $\sigma \in \{-1, +1\}^{D_a}$. For free boundary conditions the Hamiltonian is

$$H_{a,D}^{\lambda, \text{free}}(\sigma) = -J \sum_{\substack{\{u,v\} \subset D_a \\ |u-v|=a}} \sigma_u \sigma_v - \lambda \sum_{\substack{\emptyset \neq A \subset \mathbb{Z}^2 \text{ finite} \\ aA \subset D_a}} V(A) \prod_{x \in A} \sigma_{ax}. \quad (1.1)$$

Every nearest-neighbor pair and every interaction set is unordered and counted once. For $b \in \{+, -\}$ extend σ to $a\mathbb{Z}^2 \setminus D_a$ by the constant sign b . In (1.1), replace the containment condition $\{u, v\} \subset D_a$ by $\{u, v\} \cap D_a \neq \emptyset$, retaining $|u - v| = a$, and replace $aA \subset D_a$ by $aA \cap D_a \neq \emptyset$. Terms entirely outside the domain are omitted. These sums are finite by finite range. The Gibbs weight is proportional to $\exp(-\beta H)$, with no magnetic field.

Set

$$K_c = \frac{1}{2} \log(1 + \sqrt{2}), \quad \beta_0 = K_c/J.$$

For an interior point x , let $[x]_a = a[x/a]$, with the floor taken coordinatewise. Once $\beta_c(\lambda)$ is specified, write $\langle \cdot \rangle_{a,D}^{\lambda, b}$ for the Gibbs expectation at that inverse temperature, and define

$$S_a(x) = a^{-1/8} \sigma_{[x]_a}, \quad (1.2)$$

$$E_{a,j}(x) = a^{-1} \left(\sigma_{[x]_a} \sigma_{[x]_a + ae_j} - \left\langle \sigma_{[x]_a} \sigma_{[x]_a + ae_j} \right\rangle_{a,D}^{\lambda, b} \right), \quad j \in \{1, 2\}. \quad (1.3)$$

For fixed interior points these bonds lie in D_a when a is small.

Let $\mathcal{C}_D^b(\mathbf{x}; \mathbf{y})$ be the corresponding nearest-neighbor limit at $\lambda = 0$, $\beta = \beta_0$, with the same centering convention. The established primary-field results give these limits, their independence of the energy-direction labels, and their conformal covariance. Their normalization is fixed by this lattice definition, not by a separate choice of continuum constants. Write $\mathcal{C}_{\mathbb{C}}$ for the plane comparison functions.

Theorem 1.1 (Conformal universality). *For each J and potential V as above, there are $\lambda_0 > 0$ and real functions $\beta_c, Z_\sigma, Z_\epsilon$ on $(-\lambda_0, \lambda_0)$, continuous at zero, such that*

$$\beta_c(\lambda) > 0, \quad \beta_c(0) = \beta_0, \quad Z_\sigma(0) = Z_\epsilon(0) = 1, \quad Z_\sigma(\lambda) Z_\epsilon(\lambda) \neq 0.$$

For every fixed $|\lambda| < \lambda_0$, every allowed domain D , every $b \in \{\text{free}, +, -\}$, all integers $n, m \geq 0$ with $n + m \geq 1$, all pairwise distinct interior points $x_1, \dots, x_n, y_1, \dots, y_m$, and all $j_1, \dots, j_m \in \{1, 2\}$, the full-sequence limit is

$$\lim_{a \downarrow 0} \left\langle \prod_{i=1}^n S_a(x_i) \prod_{k=1}^m E_{a,j_k}(y_k) \right\rangle_{a,D}^{\lambda, b} = Z_\sigma(\lambda)^n Z_\epsilon(\lambda)^m \mathcal{C}_D^b(\mathbf{x}; \mathbf{y}). \quad (1.4)$$

The same functions and coupling interval apply to every such choice.

At each fixed mesh, the infinite-volume limit along periodic square boxes exists for all local observables. Using that state first, and then taking $a \downarrow 0$, gives (1.4) with D replaced by $\mathbb{C}$. In particular its two-spin limit is a nonzero constant times $|x - y|^{-1/4}$.

If $\phi : D \rightarrow D'$ is a conformal bijection between allowed domains, the limiting correlation in D is its counterpart at the image points in D' , multiplied by

$$\prod_{i=1}^n |\phi'(x_i)|^{1/8} \prod_{k=1}^m |\phi'(y_k)|. \quad (1.5)$$

The boundary type is transported unchanged.

The branch in Theorem 1.1 is constructed in the proof. Its critical nature in the stated sense follows from the massless correlation limits; it is not a hypothesis. Neither the interaction strength nor its shape depends on a . The theorem makes no assertion about coincident or boundary insertions, uniform convergence rates in λ , or interfaces.

1.1 Historical context and the universality prediction

The exact solution of the nearest-neighbor Ising model made its critical behavior a central example in statistical mechanics. Onsager determined the square-lattice free energy and its singularity [27]; Yang obtained the spontaneous magnetization [31]. The conformal-field-theory framework of Belavin, Polyakov, and Zamolodchikov [4] organizes the critical spin and energy fields by their dimensions $1/8$ and 1 . The universality question asks whether the same continuum correlations persist when a microscopic interaction destroys the exact nearest-neighbor structure. In Theorem 1.1, the perturbation remains fixed as the lattice mesh tends to zero.

For the critical reference model, discrete complex analysis supplies a rigorous route to conformal covariance. Chelkak and Smirnov developed uniform discrete analytic estimates on isoradial graphs [12] and proved convergence of critical Ising fermionic observables in that setting [13]. Hongler and Smirnov computed the one-point energy-density limit for fixed and free boundary conditions [25]. Chelkak, Hongler, and Izyurov established bulk spin-correlation convergence using discrete holomorphic spinors [10], and subsequently treated mixed primary fields in finitely connected domains with fixed, free, or mixed boundary arcs [11]. Dubédat's exact bosonization relates squared spin correlations to neutral-charge expressions [15]. The bosonization theorem of Bayraktaroğlu, Izyurov, Virtanen, and Webb [3] provides a modern formulation for primary fields in multiply connected domains. These reference results identify the continuum functions used here. Their exact hypotheses, and the particular homogeneous spin formulas needed for the radial argument, are recorded in Section 2.

Substantial universality results are also known beyond the nearest-neighbor model. Giuliani, Greenblatt, and Mastropietro [22, Theorems 1.1–1.2] proved the scaling limit of truncated energy correlations for small, finite-range, rotation-invariant pair perturbations of either sign. Their construction first takes the thermodynamic limit at a noncritical temperature depending on the mesh, then takes the scaling limit, including the zero limiting-mass case. The interacting Grassmann representation and constructive fermionic renormalization underlying that work are important predecessors of rigorous perturbative Ising universality, building on the Pinson–Spencer approach described in [22].

Antinucci, Giuliani, and Greenblatt [1, Theorem 1.1 and Corollary 1.2] proved energy universality on cylinders with periodic and free sides, with the half-plane obtained by exhaustion. Their interaction already includes finite-range even multispin terms of either sign; their energy is centered

by its actual interacting-domain expectation. They construct analytic critical parameters and amplitudes and control the boundary corrections, including the cancellation of a marginal boundary term. Thus signed multispin interactions and actual-domain energy centering are established features of that earlier energy theory. Cava, Giuliani, and Greenblatt [9, Theorem 1.1] subsequently proved the critical multipoint boundary-spin limit in the free half-plane, with its multiplicative amplitude and the same Pfaffian limit as the nearest-neighbor model. Those boundary spins have dimension $1/2$; the bulk spins in Theorem 1.1 have dimension $1/8$.

Giuliani’s formulation of the universality program [21, Section 2 and Open Problems 1–2] asks for mixed spin and energy limits with two field normalizations under weak even finite-range perturbations. Theorem 1.1 establishes this prediction for the square-symmetric model in smooth simply connected domains with free or either fixed boundary, and in the plane. It treats all separated mixed bulk configurations with one temperature branch and one pair of amplitudes. The broader program also includes other topologies and boundary observables; those are outside the statement proved here.

1.2 Obstacles and main ideas

Write $\mathbb{E}_0$ for expectation in the critical nearest-neighbor reference law of the geometry under consideration. Our first task is a uniform estimate for conditional averaging in the plane. Let F be a bounded function of the spins in a square of radius s . Conditioning on a square contour at radius $R \gg s$ gives a function of the cutting spins. We show that, after subtracting its reference mean, this function has essentially one leading component in each spin-flip sector: the spin component for odd F , and the energy component for even F . A distant fixed-plus disk detects the coefficient of that component. The remainder decays faster as R/s grows. These estimates must hold uniformly over all bounded F , including functions of arbitrarily many microscopic spins.

This uniformity is the first obstacle. Convergence of each fixed list of primary fields gives information on finitely many prescribed spin products. The local interaction functions created by a change of scale need not belong to any such fixed list. The fermionic approach to energy universality uses local representations of its observables, whereas bulk spin insertions introduce nonlocal defects; see [22, 1]. We instead use local functions of the physical spins throughout.

The companion *Buffered comparison and stopping-band resolution in critical Ising* [28] supplies comparison and stopping estimates for the approximation step. Using these estimates, we show that for every prescribed error, the effect of F on all distant bounded tests can be replaced by a finite signed combination of reference laws with deterministic signed contours pinned. The total absolute coefficient sum is at most $\|F\|_\infty$. The approximation follows by conditioning on a separating contour and sorting a finite stopping exploration into finitely many classes. When an outer continuation is changed, its partition-function ratio provides the same bounded weight for each inner alternative; this carries the comparison through the unchanged collar.

We then replace the signed pins by normalized weights $e^X/\mathbb{E}_0 e^X$, where X is a weighted sum of reference spins in thin tubes around the contours. The pin-to-field comparison uses the companion’s ordered transmission estimate and continuity of reference correlations. The reverse comparison adapts the cluster-necklace and ghost-attachment argument of Camia, Jiang, and Newman [8]. Magnetization moment estimates [7] justify these exponentials and successive replacements at both ends of a pairing. These comparison arguments all take place in the critical nearest-neighbor model.

Reflection positivity places the limiting soft weights in a radial Hilbert space. Their norms are bounded independently of their profiles, so the exact spin formulas give operator bounds for the whole finite approximating family. The low spectrum contains the spin line of dimension $1/8$ and the energy line of dimension 1. In the even, quarter-turn invariant sector, a finite charge identity

cancels the possible contribution of dimension two, leaving a remainder of order $(s/R)^3$. This extra power is decisive: summing local interactions over a two-dimensional block costs two powers of the scale ratio. The remaining bulk directions therefore contract. Separate half-plane estimates give the contraction needed for terms meeting the physical boundary.

The second task is to turn conditional averaging into an exact change of interactions. In a finite volume, represent an interaction as a list of bounded local functions and expand its exponential in lists of occurrences. An occurrence is conditionally averaged only when its support fits inside its cutting square and an enlarged neighborhood of its support is disjoint from those of all other occurrences. Overlapping occurrences are retained together. The conditional Markov property preserves the reference integral of each list. Taking the pointwise logarithm gives a new interaction H' and a scalar c . If H is the sum of the input functions, then

$$\mathbb{E}_0 e^H = e^c \mathbb{E}_0 e^{H'}.$$

The connected logarithm and tree estimates give an analytic map with exponential locality. We use the hard-core Mayer and tree-graph methods [18, 29], together with the occurrence-counting argument in Section 5. Exact local coordinate maps and contracted remainders also occur in constructive renormalization [5]; the reference kernels and list transformation used here are proved directly.

The Kadanoff–Wilson scale-flow picture explains universality by the decay of irrelevant interactions [26, 30]. Here the linear estimate leaves one expanding coordinate, the thermal one. Solving its recurrence backward while solving the contracting coordinates forward expresses the thermal coordinate as an analytic function of the contracting coordinates. This graph consists of interactions that decay under repeated changes of scale. Related sequence-space constructions appear in [2]; our contraction argument is Lemma 6.1. The microscopic temperature direction crosses this graph transversely and selects $\beta_c(\lambda)$. Where a cutting square lies in the interior of a physical domain, its conditional kernel agrees with the plane kernel. Boundary contraction controls the remaining edge terms; exponential locality controls the discrepancies on periodic boxes. These comparisons carry the same temperature branch to all the geometries in the theorem.

Finally we insert finitely many marked spin and energy observables in the same exact partition identity. Each single source approaches a conditional average of its reference field, multiplied by one domain-independent amplitude. We stop at a fixed small physical radius, while the source neighborhoods are still disjoint. Connected interactions between distinct sources then vanish, using exponential locality together with the decay of the background or of a one-source remainder.

At this stopping scale the total residual background in a bounded domain tends to zero, so the surviving source fields can be averaged in the critical reference measure. This measure retains their mixed correlations. Conditional independence in the disjoint cutting squares recovers every subset moment of the microscopic reference fields. Removing the energy singleton coefficients from the logarithmic generating function gives centering by the actual interacting state. An additional contraction for unscaled observables constructs the periodic thermodynamic state before the continuum limit and compares its residual local laws with the reference laws. The resulting massless correlations establish the critical behavior of the selected branch.

1.3 Proof structure

Section 2 fixes the reference conventions and states the exact equilibrium interfaces and primary-field inputs. Section 3 proves the finite soft-field approximation, including its uniformity over bounded tests and its moment bounds. Section 4 constructs the radial space, proves the charge cancellation and low-spectrum estimates, and derives the uniform bulk and boundary conditional-averaging

bounds. Section 5 builds the exact analytic map and separates the thermal, marked, and boundary directions. Section 6 constructs the temperature branch and compares its plane trajectory with physical domains and periodic boxes. Section 7 tracks the marked sources, performs actual-state energy centering, establishes the order of plane limits, and completes Theorem 1.1.

The order of choices is a large fixed scale ratio, a sufficiently large initial lattice scale, and then a sufficiently small coupling interval. No quantitative rate of primary-field or interface convergence is required. Only the terminal physical scale depends on the geometry of a fixed domain.

2 Reference model and equilibrium inputs

2.1 Reference conventions

All conditional averaging below uses the *critical nearest-neighbor* measure at bond coupling K_c , in the plane or with the indicated physical boundary condition. We write μ^0 for such a reference law and identify which geometry is meant whenever it matters. For a finite cut, the conditional law is determined by its cutting spins and physical boundary data; it is not a conditional law of the interacting system.

After rescaling a local chart, its fine mesh is denoted by δ . For the renormalization construction we return to lattice-unit lengths. There $B_r(z)$ is a max-norm square centered at z , and $P_{r,z}F$ is the conditional expectation of F given the exterior and the cutting contour of $B_r(z)$. A consistent rounding convention includes the cutting vertices in the retained exterior. The resulting function depends only on those vertices, together with the prescribed physical boundary. Nested cuts compose by conditional expectation. Bounded changes in lattice rounding are absorbed in fixed buffer margins. Write $P_r = P_{r,0}$.

The microscopic scalar energy used for the distinguished direction is

$$e(z) = \frac{1}{4} \sum_{v: |v-z|=1} \sigma_z \sigma_v. \quad (2.1)$$

Each unoriented bond occurs twice in this sum over z , so $\sum_z e(z)$ is half the total bond-product sum. The source fields are $\mathcal{O}_1 = e$ and $\mathcal{O}_{1/8} = \sigma$, with the subscript indicating scaling dimension. Individual directed bonds are retained in the target; their common normalization will follow from square symmetry, not from changing the observable in (1.3).

Supremum norms of local functions are actual suprema over their available spins. Background interactions are considered modulo constants, represented by subtracting the mean under independent uniform available spins. This projection costs at most a factor two in the supremum norm. Source scalar terms will be recorded separately. All constants may depend on fixed buffer geometry. Once the scale ratio is chosen, analytic constants may also depend on that ratio and the fixed potential; constants for source jets may depend on the finite number of marks.

2.2 Equilibrium inputs

Every measure in this section is ferromagnetic. Except in the finite-graph statement below, its nearest-neighbor coupling is K_c . External fields introduced here are auxiliary fields in this reference model. None of the comparison inequalities is asserted for the perturbed Hamiltonian of Theorem 1.1.

We state the particular consequences of the established reference theory that will be used. The pointwise, ordered-transfer, and stopping statements are proved in the companion article *Buffered comparison and stopping-band resolution in critical Ising* [28]. We state every interface used here,

including its restrictions on common pins and unchanged continuations. The pointwise comparison uses the external-field inequality of Ding–Song–Sun [14, Corollary 1.3].

Lemma 2.1 (Pointwise comparison with common pins). *Let G be a finite graph with nonnegative pair couplings and arbitrary finite real fields. Let I, J, F be disjoint vertex sets, with I, J nonempty, and prescribe arbitrary common spin values on F . Form the graph obtained by deleting F and contracting I and J separately; parallel couplings are added. Let q_0 be the probability that the two contracted vertices are FK-connected in the zero-field, cluster-weight 2 measure on this graph. For any configurations a on I and b, b' on J ,*

$$e^{-4 \operatorname{arctanh} q_0} \leq \frac{\mathbb{P}(\sigma_I = a \mid \sigma_J = b, \sigma_F)}{\mathbb{P}(\sigma_I = a \mid \sigma_J = b', \sigma_F)} \leq e^{4 \operatorname{arctanh} q_0}. \quad (2.2)$$

Either conditional law may be replaced by a mixture of such laws. All external fields, including those produced by the common pins, are set to zero in the comparator defining q_0 .

This is [28, Lemma 2.2]. Common pins are deleted vertices in the comparator; they are not additional FK wires. This distinction will allow the same estimate in the presence of many signed pins.

Corollary 2.2 (Buffered and shrinking-patch comparison). *Fix $c > 0$. If $I \subset B_R(z)$ and $J \subset B_{(1+c)R}(z)^c$, their conditional density ratios in Lemma 2.1 lie in $[C_c^{-1}, C_c]$, uniformly in the mesh, common pins, and fields. There are $C < \infty$ and $\theta > 0$ such that, if $I \subset B_w(z)$, $J \subset B_R(z)^c$, and $R \geq 8w \geq 8$ in lattice units, the absolute log density ratio is at most*

$$C(w/R)^\theta. \quad (2.3)$$

No unpinned neighborhood of I is required.

The buffered bound holds for fixed polygonal separating collars with positive relative width, and for their intersections with a free or fixed physical boundary. It also compares different continuations or volumes when the reference interaction agrees in the separating chart. The same assertions hold for mixtures and after a common conditional randomization of the local spins. Finitely many shrinking patches can be changed successively.

Proof. For square collars this is [28, Corollaries 2.5–2.6]. Its proof uses the critical FK-Ising RSW estimate with arbitrary boundary wiring [16]. A finite collection of rectangular dual crossings supplies a separating circuit with probability bounded below. Repeated disjoint annular tests give $q_0 \leq C(w/R)^\theta$. In a clipped chart restore the missing lattice vertices and edges when bounding q_0 from above, and then apply the same tests in the restored collar. Common fixed boundary spins have already been deleted. Polygonal collars use finitely many such rectangular tests. Conditioning on a farther contour represents each continuation as a mixture of the same local kernels. These observations give the stated extensions of the square version. $\square$

For clarity, we give the geometry needed in the ordered-transfer statement. A cut J separates a source S_0 from a target T in the graph of unpinned sites. At lattice scales $1 \ll w \leq s$, call the cut *admissible* if it is partitioned into patches b_i , with straight counting segments $L_i \subset J$ and centers z_i , such that the following hold with fixed positive comparison constants. Each $b_i \cup L_i$ lies in a square of radius $\ell_i \asymp w$; $|L_i| \asymp \ell_i$; the square $B_{16\ell_i}(z_i)$ is an unpinned lattice chart disjoint from the source; the L_i have bounded overlap; and T is a distance comparable to s away, with $16\ell_i$ smaller than that distance. The patch estimate must be available up to a fixed multiple of s . A square contour in a

pin-free collar of width comparable to w satisfies these conditions after subdivision, including at its corners. Put

$$M_J = s^{-7/8} \sum_{x \in J} \sigma_x.$$

Corollary 2.3 (Ordered changes with a common continuation). *Suppose $\mu^- \preceq \mu^+$ are two ordered laws, possibly on different volumes or with different pins. Assume there is a common unpinned source contour S_0 such that, given its spins, both laws have the same critical nearest-neighbor continuation toward T . Apart from the common pinning data, this continuation has zero field. All changed data lie on the source side of S_0 . For an admissible intermediate cut J , there are $C < \infty$ and $\theta \in (0, 1/8)$ such that*

$$\|\mu_T^+ - \mu_T^-\|_{\text{TV}} \leq C(w/s)^{-7/8+\theta} (\mathbb{E}_{\mu^+} M_J - \mathbb{E}_{\mu^-} M_J). \quad (2.4)$$

Here total variation is the supremum of event-probability differences. The constants depend on the fixed chart and admissibility constants, not on the source values or their number. At a fixed positive w/s , vanishing rescaled magnetic difference implies vanishing target total variation. Random ordered source configurations are allowed.

This is [28, Corollary 4.12, with Definition 4.1 and Corollary 4.11]. Fields or additional changes before S_0 are allowed by first coupling the induced source distributions by order and then using the common continuation. The target-side continuation has no auxiliary field, and its pinning data are identical in the two laws.

Theorem 2.4 (Outward stopping-band transfer). *Work at mesh δ in a fixed square chart with homogeneous fixed sign on its outer wall. Fix nested squares $Q_u \subseteq Q_v$ and a target T outside Q_v and inside the wall, with positive clearances from both. Reserve a slightly larger band window, also separated from T and the wall. The levels u, v may be chosen generically in prescribed separated subintervals of this window.*

Initially reveal Q_u and its cutting contour. There is a finite adaptive spin exploration through the band, with full transcript $\mathcal{F}^\delta$, such that the unexamined sites, conditional on each realizable transcript, have the ordinary Gibbs law with the examined spins fixed. Given $\epsilon, \eta > 0$, this observation has a descriptor

$$Z^\delta \in \{0, 1, \dots, m\}$$

computed from band spins, with the following properties for all sufficiently small δ :

1. $\mathbb{P}(Z^\delta = 0) \leq \epsilon$ in the fixed-wall reference law.
2. For every $i \geq 1$ there is a deterministic regular signed Jordan barrier Γ_i , with finitely many sign arcs, on the fresh side of the target-facing revealed boundary. It separates the reveal from T and has a positive clearance from T and the distant wall. For every realizable transcript with $Z^\delta = i$, adding the plus barrier pins and then the minus barrier pins changes the target law by a sum of at most η in total variation.
3. After these additions, the target-side law is a single law $Q_{i,\delta}$, determined by Γ_i and the unchanged outer wall, independent of the particular transcript in bin i .

The finite list, all its positive clearances, and m are chosen before the mesh threshold, depending only on the geometry and ϵ, η . They work simultaneously for all bounded target tests, including mesh-dependent tests. For a finite list of bands, the descriptors may be chosen with converging joint laws. If the band-spin distribution under another law has density at most C relative to the reference distribution, its exceptional probability is at most $C\epsilon$.

The statement is the outward part of [28, Theorem 9.1 and Lemma 7.2]. In particular, for $|H_\delta| \leq 1$, on every good transcript,

$$\left| \mathbb{E}[H_\delta \mid \mathcal{F}^\delta] - \mathbb{E}_{Q_{Z^\delta, \delta}} H_\delta \right| \leq 2\eta. \quad (2.5)$$

The barriers can be polygonal or regular smooth contours. They need not lie between the two numerical levels of an individual band; the reserved window accommodates their offsets and tip caps. The last assertion transfers the *exceptional probability*. A change of target-side continuation requires an additional buffered comparison; this is carried out in Section 3.

Proposition 2.5 (Static critical-field inputs). *The following facts concern only the critical nearest-neighbor model.*

1. *For bounded finitely connected domains without singleton boundary components and with finitely many nondegenerate fixed-plus, fixed-minus, or free boundary arcs, separated bulk spin and energy correlations converge with dimensions $1/8$ and 1 . Discrete domains converge in the Caratheodory sense, with eventually matching boundary-component and arc counts and converging boundary marks. Convergence is uniform on compact separated configurations. The limits are conformally covariant and continuous under these marked-domain limits. The energy convention initially subtracts its plane mean; expanding the finitely many subtractions gives the same assertion for domain-centered energies. The two bond directions have the same scalar limit. Homogeneous half-plane spin limits and full-plane spin limits are included, with odd plane correlations zero.*
2. *Normalize the continuum spin so that its plane two-point function is $|z - w|^{-1/4}$. Write G_n for its n -point function. For even n ,*

$$G_{n, \mathbb{C}}(z_1, \dots, z_n)^2 = 2^{-n/2} \sum_{\substack{\mu \in \{-1, 1\}^n \\ \sum_i \mu_i = 0}} \prod_{i < j} |z_i - z_j|^{\mu_i \mu_j / 2}. \quad (2.6)$$

For z_i in the upper half-plane, put $q_{ij} = |(z_i - z_j)/(z_i - \bar{z}_j)|$. Then

$$G_{n, \mathbb{H}}^+(z)^2 = 2^{-n/2} \prod_i (2 \operatorname{Im} z_i)^{-1/4} \sum_{\mu \in \{-1, 1\}^n} \prod_{i < j} q_{ij}^{\mu_i \mu_j / 2}, \quad (2.7)$$

$$G_{n, \mathbb{H}}^{\text{free}}(z)^2 = c_n \prod_i (2 \operatorname{Im} z_i)^{-1/4} \sum_{\mu \in \{-1, 1\}^n} \left(\prod_i \mu_i \right) \prod_{i < j} q_{ij}^{\mu_i \mu_j / 2}, \quad n \text{ even}. \quad (2.8)$$

Here $c_n = (-1)^{n/2} 2^{-n/2}$ for even n , including $c_0 = 1$. The plus formula holds also for odd n ; minus boundary is obtained by spin flip. These functions inherit reflection positivity from the lattice and satisfy spherical covariance in the plane. The plus-disk one-point functions of the spin and the plane-centered scalar energy are nonzero.

3. *For homogeneous plus, minus, or free reference laws and for the plane, the rescaled two-point bound is*

$$\delta^{-1/4} |\mathbb{E} \sigma_z \sigma_{z'}| \leq C(|z - z'| \vee \delta)^{-1/4}$$

on buffered compact sets. More generally, if $z_1, \dots, z_n$ lie in a fixed compact interior set and ρ_i is the lesser of a fixed buffer radius and one quarter of the distance to the other insertions, then

$$\left| \mathbb{E} \prod_i \sigma_{z_i} \right| \leq C_n \prod_i \left(\frac{\delta}{\rho_i \vee \delta} \right)^{1/8}. \quad (2.9)$$

Repeated sites are allowed with the factor capped at one. The estimate uses ordinary FK wiring and homogeneous cluster coloring; prescribed mixed boundary signs are handled by buffered density comparison. For bounded compactly supported piecewise regular profiles h , the variables $X_\delta(h) = \delta^{15/8} \sum_z h(z) \sigma_z$ have uniform exponential moments of every fixed real order under homogeneous reference laws on buffered compact sets. Positive-profile bounds follow from GHS; signed profiles follow by splitting the signs and applying Cauchy–Schwarz. These moment inputs also hold in homogeneous half-plane references by local comparison.

4. In a fixed square with free boundary and zero field, for every $\epsilon > 0$ there are $K < \infty$ and $c > 0$ such that, for all sufficiently small δ , with probability at least $1 - \epsilon$ a nearest-neighbor vertex circuit surrounds the concentric square of one third the side length. The circuit is contained in the union of at most K FK clusters, each of cardinality at least $c\delta^{-15/8}$; successive clusters may be separated by one nearest-neighbor step. For a uniform positive field $h\delta^{15/8}$, conditional on internal FK bonds, a cluster C is attached to the plus ghost with probability

$$\tanh(h\delta^{15/8}|C|), \quad (2.10)$$

independently for distinct internal clusters. Internal bonds dominate their zero-field law. Consequently, at fixed physical square size, the probability of a plus-spin circuit tends to one by choosing a sufficiently large finite h and then a sufficiently small mesh.

The bounded-domain convergence is [11, Theorem 1.3 and Definitions 3.1–3.2]; its Theorem 7.1 gives (2.7). The plane neutral-charge form is [15, Theorem 4], with the displayed normalization agreeing with the half-plane formula as the boundary recedes. The homogeneous free formula also follows directly from [3, Theorem 1.1 and Definition 2.3]: the constant boundary shift $\pi/\sqrt{2}$ changes each bosonic cosine into a negative sine, whose charge expansion has coefficient $i\mu_i/\sqrt{2}$ for charge $\mu_i \in \{-1, 1\}$. Multiplication over an even number of insertions gives $c_n \prod_i \mu_i$ in (2.8). The half-plane Green function $-\log q_{ij}$ and its regularized diagonal $\log(2 \operatorname{Im} z_i)$ supply the displayed pair and height factors. The bounded-domain theorem and exhausting homogeneous domains supply half-plane references; changing a distant wall is controlled by Corollary 2.2. The nonzero plane-centered energy response in a fixed-plus disk is also explicit in [25, Theorem 1.1 and Corollary 1.2]. No rate of primary-field convergence is used.

Continuity under the stated continuum marked-domain limits follows from the same discrete convergence theorem. Given a nondegenerate sequence $D_j \rightarrow D$, choose lattice approximations of D_j at meshes $\delta_j < 1/j$ for which the rescaled correlations differ from their continuum values by less than $1/j$. Choose them sufficiently close to D_j that the diagonal sequence converges to D with the required component and arc counts and boundary marks. Applying the discrete convergence theorem to this sequence proves continuity. Allowing the marked insertion configurations to vary in a compact separated set gives its locally uniform version. This deduction keeps the prescribed topology and nondegenerate limiting boundary arcs.

For the third item, the FK one-arm estimate and product bound are recorded in [28, Lemmas 3.1–3.3]. The exponential-moment input is [7, Section 3.1, Proposition 3.5, Theorem 3.6, Corollaries 3.7–3.8]; the passage from these inputs to the integrated limits used here is proved in Lemma 3.4. The last item is [8, arXiv version 4, Lemmas 4 and 8 and the proof of Lemma 9]. Indeed its success probability is bounded below by $(1 - \epsilon) \tanh(ch)^K$, after an arbitrary initial reduction of ϵ . The event may be formulated using connected subsets of the required size, so merging clusters preserves the geometric event.

We use finite-volume positive association and stochastic ordering [19, 24], the GHS inequality [23], the Edwards–Sokal coupling [17], and Euclidean reflection positivity [20] only under the hypotheses

stated above. Whenever an infinite-volume reference is used, the finite comparison is first applied inside a chart and then its outer wall is sent to infinity. All its constants are independent of that wall.

3 Finite soft fields for arbitrary bounded tests

All measures in this section are nearest-neighbor Ising measures at K_c . In particular, a magnetic weight introduced here is an auxiliary weight in a reference measure; it is not an interaction in the model of Theorem 1.1. We write δ for the mesh after a buffer geometry has been rescaled to fixed size. A contour includes the lattice sites needed for its nearest-neighbor Markov property. Changes of this convention by a bounded number of lattice layers will not affect any positive clearance below.

The purpose of the section is to approximate the *transmitted effect* of a bounded observable. The observable itself need not have a scaling limit, nor need it be approximable pointwise by a function of finitely many continuum fields.

3.1 Reference charts and changes of continuation

The reference geometries are the plane and a half-plane with homogeneous fixed or free boundary. In a half-plane chart we also allow lattice domains whose physical boundaries, on successively larger windows, converge to the line as single C^1 graphs. The boundary prescription is the same throughout the chart. Such a chart admits reference measures with the usual bulk spin convergence. Indeed, first fix a large window and continue its central boundary graph, by a cutoff, to the limiting line outside a smaller window. Close the resulting domain with regular distant walls of the same boundary type. The boundary parametrizations and any marked arcs converge, so Proposition 2.5 applies. The walls can subsequently be sent to infinity, or chosen to increase sufficiently slowly with decreasing mesh.

For clarity, the lattice approximations needed in this construction are compatible with site-induced domains. A compact smooth boundary has a finite cover by graph patches transverse to one of the coordinate axes. Within a smaller such patch, a nearby interior site can move in the transverse coordinate into a region with positive interior clearance; exterior sites have the corresponding property on their side. Lattice paths approximate paths in each buffered region. This excludes alternating inside/outside diagonal pockets and gives the requisite local connectivity and boundary convergence. It also allows regular inner charts to be chosen away from the ends of a graph. The finitely connected versions of Proposition 2.5 will be used when a pinned contour forms an inner boundary.

The dependence on a distant wall vanishes on fixed compact sets by Corollary 2.2, iterated through annuli. Thus these references give the plane or half-plane spin formulas. The same conclusion for unbounded smeared tests follows from the moment bounds proved below. All reference changes made before a fixed buffer limit have fixed positive clearances.

We shall repeatedly use the following elementary consequence of the pointwise ratio estimate. It is useful to distinguish it from the exceptional-probability assertion in Theorem 2.4.

Lemma 3.1 (Transport through an unchanged collar). *Let T be a separating contour and W a farther contour, with a fixed positive buffer between them. Fix two outer Gibbs continuations, independently of the data on the inner side of T . They agree on the graph, couplings, and fields between T and W , and all their differences are on or beyond W . For any fixed data on the inner side of T , the corresponding laws on T differ by a positive tilt w satisfying*

$$\frac{\sup w}{\inf w} \leq K, \quad (3.1)$$

where K depends only on the buffer geometry. The same weight w works for every change of the inner data. Common pins and arbitrary common finite fields are permitted wherever Lemma 2.1 permits them.

If two laws ν_1, ν_2 on T are compared in one continuation, their laws in the other satisfy

$$\|\nu_1^w - \nu_2^w\|_{\text{TV}} \leq 2K \|\nu_1 - \nu_2\|_{\text{TV}}, \quad d\nu_i^w = \frac{w d\nu_i}{\nu_i(w)}. \quad (3.2)$$

In the destination continuation, both tilted laws use the same further Gibbs kernel given T . Applying that kernel preserves the bound. Remote data may first be conditioned on, or integrated out in the outer Gibbs weights before the inner data are varied.

Proof. For a contour configuration τ , let $Z_1^{\text{out}}(\tau)$ and $Z_2^{\text{out}}(\tau)$ be the two outer continuation weights, including any summation over remote data. Their ratio $w(\tau) = Z_2^{\text{out}}(\tau)/Z_1^{\text{out}}(\tau)$ is independent of every inner datum. For a fixed configuration at W , Lemma 2.1 bounds the oscillation of this ratio over τ ; normalization of the contour marginal does not change its supremum-to-infimum ratio. Conditioning at W and mixing proves (3.1) for an arbitrary continuation. This argument also applies to a different outer domain, since the graph in the collar is unchanged.

Rescale w so that $1 \leq w \leq K$, and put $d = \|\nu_1 - \nu_2\|_{\text{TV}}$. For an event A ,

$$\left| \frac{\nu_1(w\mathbf{1}_A)}{\nu_1(w)} - \frac{\nu_2(w\mathbf{1}_A)}{\nu_2(w)} \right| \leq Kd + |\nu_1(w) - \nu_2(w)| \leq 2Kd.$$

The bounds on the integrals follow from the layer-cake formula for nonnegative functions bounded by K . Taking the supremum over A proves (3.2). Conditional on T , the actual outer Gibbs kernel is the same for both inner laws, so applying that kernel cannot increase total variation. $\square$

Figure 1 places this unchanged collar outside the window reserved for a stopping barrier. The contour T carries the common tilt for every barrier lying in that window.

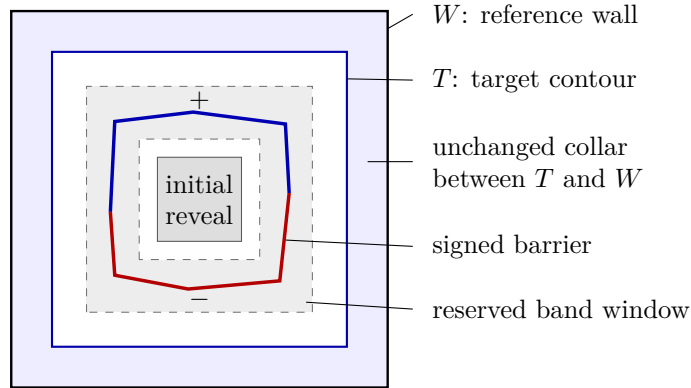


Figure 1: One buffered source square, drawn schematically. Dashed squares delimit the reserved band window; the numerical stopping levels are omitted. The signed barrier separates the initial reveal from T , with room for offsets and tip caps inside the window. Changing the continuation beyond W multiplies both laws on T by the same bounded-oscillation weight.

3.2 Deterministic barriers and soft fields

For a bounded, piecewise regular profile h with compact interior support, set

$$X_{\delta,h} = \delta^{15/8} \sum_z h(z) \sigma_z, \quad U_{\delta,h} = \frac{\exp(X_{\delta,h})}{\mathbb{E} \exp(X_{\delta,h})}. \quad (3.3)$$

The expectation in the denominator is in the chosen homogeneous reference law. Profiles may be positive or negative. Sampling their fixed regular pieces on the lattice, with bounded lattice rounding of their boundaries, is understood.

Lemma 3.2 (Soft realization of a signed barrier). *Fix a regular signed Jordan contour with finitely many sign arcs in an interior square, a buffered outer target contour, and a more distant square wall with a constant prescribed sign. Given $\eta > 0$, there is a bounded piecewise regular profile h , supported in arbitrarily small prescribed neighborhoods of the signed contour, such that, for all sufficiently small δ , the law on the target with the contour pinned and the law tilted by $U_{\delta,h}$ have total variation less than η . The contour pins are not imposed in the latter law. The profile and all its nonzero tube widths are fixed before δ tends to zero. The conclusion transfers through any unchanged outer buffer as in Lemma 3.1, and can be applied successively to finitely many disjoint buffered squares.*

Proof. We compare the pinned law, the law with both pins and fields, and the law with fields alone. All error budgets in this proof may first be divided by the finite transport constants in Lemma 3.1.

Fields in a pinned contour. Delete small neighborhoods of the finitely many sign switches. Around the remaining cores of the plus and minus arcs choose thin regular tubes, with the matching field sign, and put zero field elsewhere. The field has magnitude at least $h_0 \delta^{15/8}$ on each tube core; the finite number h_0 will be chosen last. Fields on the inside of the full pinned contour have no effect on its exterior.

Outside the contour, both the pinned law and the pinned-plus-field law are between the following ordered laws. For an upper law enlarge the plus arcs into fully plus-pinned strips and omit the negative fields; for a lower law use minus-pinned strips and omit the positive fields. All original contour pins remain. Strip shapes can be chosen as regular bulges attached to just the corresponding sign arcs. Their ends stay away from the switches, and the outer wall is unchanged. After integrating irrelevant components, the target components are regular marked domains, possibly annular. They converge back to the original marked domain as the strip widths tend to zero.

Choose a square source contour S_0 beyond all strips and an intermediate square cut J between S_0 and the target, with fixed pin-free collars. At fixed positive strip widths, Proposition 2.5 gives convergence of the rescaled one-point functions on J . Their limits converge to those in the original domain when the widths shrink. They are uniformly bounded on J : its distance from every pin is positive, and the fully wired one-arm bound controls the absolute magnetization. Consequently the difference of the normalized magnetic sums on J tends to zero. In rescaled coordinates this normalization is $\delta^{7/8} \sum_{z \in J} \sigma_z$, up to fixed geometric factors.

For either of the two laws being compared, couple it in order below the common upper law. The magnetic difference is bounded by the full upper-minus-lower difference. The continuation from S_0 to the target is identical in these comparisons, and has no field or changed pin. Corollary 2.3 therefore gives an arbitrarily small target-TV distance. This conclusion is uniform in h_0 , because the extreme pinned-strip laws do not depend on the field magnitude.

Pins in a soft field. Keep the positive tube widths just chosen. Cover each arc core by finitely many patches, each contained in the central third of a square whose slightly larger buffered square is still inside the matching-sign tube. Consider first a plus patch and the comparison law with free boundary on its larger square and homogeneous field $h_0\delta^{15/8}$.

By the necklace input in Proposition 2.5, given $\xi > 0$ there are finite K and $c > 0$ such that, at sufficiently small mesh, with probability at least $1 - \xi$ the central square is surrounded by a vertex circuit covered by at most K critical FK clusters, each with at least $c\delta^{-15/8}$ vertices. Consecutive pieces may touch by a nearest-neighbor step. Equivalently, the event can be expressed using connected subsets of that minimum size; it is then increasing under addition of bonds. The internal-bond law in the positive field dominates the zero-field law. Conditional on its internal clusters, each qualifying cluster attaches to the plus ghost with probability at least $\tanh(ch_0)$. The usual conditional cluster formula, or a union bound over the at most K clusters, now gives a plus-spin circuit with probability at least

$$1 - \xi - K(1 - \tanh(ch_0)). \quad (3.4)$$

The same estimate holds for a larger matching positive field by monotonicity. Thus its failure probability can be made arbitrarily small with a sufficiently large but finite h_0 .

Across the buffer of the larger square, Lemma 2.1 compares this local law to the actual one, uniformly in all common fields and remote signs. It multiplies the failure probability by a fixed constant. Matching pins inserted earlier inside the local chart preserve the circuit event by monotonicity; other pins are covered by the same comparison. The argument for a minus patch is obtained by spin flip.

Here is the coupling that turns the circuit estimate into a pinning estimate. Couple the law before a plus patch is pinned and the law after it is pinned in stochastic order. Start at the patch and reveal, in order, the non-plus sites accessible from it by star adjacency, together with their first plus boundary. At each step use the ordered conditional distributions of the next site. This is a finite adaptive spin reveal; conditional on its revealed spins, the unobserved law is the original Gibbs continuation. A nearest-neighbor plus circuit in the lower configuration confines the exploration before it reaches the target. The separating plus boundary is plus in both configurations. The exterior conditional laws are then identical and can be sampled together. It follows that the target-TV distance is at most the lower law's circuit failure probability. This uses a stopping reveal, rather than conditioning on an arbitrarily selected random circuit. For minus pinning reverse the order.

Apply the construction to the finite patch cover, allocating the variation budget among its members. Pins still missing near sign switches lie in finitely many small disks. By Corollary 2.2, inserting these pins changes the target law by an arbitrarily small amount, uniformly in the fields and the other pins. Thus the field-only law and the pinned-plus-field law are close on the target.

The choices have the required order: first make the switch neighborhoods small; next make the strips thin but of positive width; next choose a sufficiently large finite h_0 ; finally choose the mesh small enough. The first comparison was uniform in h_0 . Combining the two comparisons proves the local statement. Lemma 3.1 transports each comparison to any allowed continuation. For finitely many disjoint squares, condition on all other pins or fields and telescope; their differences remain beyond the square's outer buffer. $\square$

3.3 Uniform replacement of bounded observables

Fix concentric buffer geometries with inner scale s and outer scale R , where R/s is a sufficiently large fixed number. An inner observable is supported in B_s , and an outer observable is supported

on or beyond $\partial B_{R/2}$. There is room for a fixed-factor buffer at either end. In a half-plane chart the center is on the limiting boundary line. Constants denoted $c, C > 0$ below depend only on these fixed support and separation factors. In particular, they do not depend on an approximation tolerance.

Lemma 3.3 (Finite soft-weight replacement). *In the preceding reference geometries, for every $\eta > 0$ there is a finite list of bounded piecewise regular profiles $h_1, \dots, h_N$, and a fine-mesh threshold, such that every inner observable F_δ with $\|F_\delta\|_\infty \leq 1$ has coefficients $c_1(\delta), \dots, c_N(\delta)$ satisfying*

$$\sum_{i=1}^N |c_i(\delta)| \leq 1, \quad \sup_{\|H_\delta\|_\infty \leq 1} \left| \mathbb{E}[F_\delta H_\delta] - \sum_{i=1}^N c_i(\delta) \mathbb{E}[U_{\delta, h_i} H_\delta] \right| < \eta. \quad (3.5)$$

The supremum is over all outer observables, including the constant one. The profiles and threshold are independent of F_δ and H_δ . The coefficients may depend on the mesh and on the observable. The analogous statement holds with the inner and outer roles reversed.

Inner replacement profiles can be supported within distance Cs of the center; outer replacement profiles can be supported where $cR \leq |z| \leq CR$. The two profile regions have a fixed relative radial separation. Each profile is supported in a compact interior set avoiding the center, with additional positive clearances that may depend on η .

More precisely, let $(F_{\alpha, \delta})_{\alpha=1}^p$ and $(H_{\beta, \delta})_{\beta=1}^q$ be finite families of inner and outer observables, all of sup norm at most one. There are finite inner and outer profile lists (h_i) and (g_j) and coefficient vectors $c_{\alpha i}(\delta), d_{\beta j}(\delta)$ such that

$$\sum_i |c_{\alpha i}(\delta)| \leq 1, \quad \sum_j |d_{\beta j}(\delta)| \leq 1, \quad (3.6)$$

$$\left| \mathbb{E}[F_{\alpha, \delta} H_{\beta, \delta}] - \sum_{i,j} c_{\alpha i}(\delta) d_{\beta j}(\delta) \mathbb{E}[U_{\delta, h_i} U_{\delta, g_j}] \right| < \eta \quad \text{for every } \alpha, \beta. \quad (3.7)$$

The same row and column vectors work throughout this matrix. Adjoin the zero profile, for which $U_{\delta, 0} = 1$, to each list. A constant observable can then be retained exactly; adjoining it to either family controls all the separate means with the same coefficients. The profile lists and mesh threshold are independent of the observables. The preceding support bounds apply to both lists. Parity and plane lattice-rotation symmetries may be imposed by averaging. The same conclusions hold along the approaching-half-plane charts described above.

Proof. We first prove the one-sided assertion. Smooth the observable by conditioning on a separating square contour at its end of the buffer. For the outer observable, a square just inside $B_{R/2}$ can be used. The conditional mean is bounded by one, and replacing the observable by that mean preserves its pairings with all tests on the other side of the cut.

Break the contour at its corners and, in a boundary chart, at its intersections with the physical boundary. Choose intersections transversely and away from corners, or include a coincident corner in the same deletion. Delete short pieces in small disks at these points. The values in a deleted piece can be assigned arbitrarily in the far conditional law, at a cost tending to zero by Corollary 2.2. More explicitly, average the original bounded contour function over the deleted spins conditional on the retained ones. For every assignment of the retained spins, compare the far law for each deleted-spin assignment to a single prescribed one. The uniform pointwise ratio bound controls the resulting pairing error. Comparing that prescribed law back to the conditional mixture costs the same bound. Thus the new pairing is again in the original reference measure, with the conditionally averaged

contour function. This operation does not assert a pointwise approximation of the observable. It also handles the vanishing displacements of intersection points in an approaching boundary chart.

The retained segments have positive distance from the physical edge and from the far test. Subdivide them into finitely many sufficiently short straight pieces, and delete much smaller gaps at the subdivision points by the same argument. Each piece now lies strictly inside its own square, and these squares have disjoint enlarged buffers. The buffers may be thin: for adjacent pieces of length ℓ separated by a gap g , one may take square half-sides just above $\ell/2$ and outer walls at $\ell/2 + g/4$, with strictly nested band and target levels in between. The stopping theorem allows any such fixed positive clearances [28, Remark 7.1]; its constants need not be uniform as g/ℓ shrinks. Choose the piece lengths after the original corner and physical-edge deletions, smaller than their positive geometric clearances. This also keeps squares on different sides disjoint. All these sizes and their number are fixed before the mesh limit. The observable, for the purpose of the desired pairings, has become a bounded function of spins in these finitely many source squares.

In each square reveal an interior square containing its source piece and apply the outward part of Theorem 2.4. Choose a separating target square contour T outside the stopping band, and a more distant constant-sign reference wall W , both still inside the square's buffer; see Figure 1. The theorem gives finitely many deterministic regular signed barriers. Off an exceptional descriptor, adding the chosen barrier changes the reference target law by the prescribed small total-variation amount, for every realizable stopping transcript.

The actual continuation can include all other source squares, their transcripts, and the physical boundary. These data lie beyond W . The law on T is therefore tilted by the common weight in Lemma 3.1, independently of the transcript and its replacement. The per-transcript TV estimate transports to this continuation. Its further law is the same given T . The exceptional probability is still small, by the buffered comparison for the band marginal. Thus both required parts of the stopping theorem transport: the exceptional-event estimate uses the band density, and the conditional target estimate uses the common target-contour tilt.

Apply these comparisons successively to the finite family of source squares. Conditional on their transcripts the remaining law is Gibbs with the revealed pins. Each square's unchanged outer collar permits the same conditional comparison, so the errors telescope. After all barriers have been inserted, they separate the initially observed spins from every far test. The law for the latter is the original reference law with just the selected deterministic signed barriers imposed; it no longer depends on the transcript behind them.

Let A_i be the good descriptor bins, with a joint index if several squares are used. The corresponding coefficient is

$$c_i(\delta) = \mathbb{E}[F_\delta \mathbf{1}_{A_i}], \quad \sum_i |c_i(\delta)| \leq \mathbb{E}[|F_\delta| \mathbf{1}_{\cup_i A_i}] \leq 1.$$

Here F_δ denotes the bounded function after the initial conditioning and deletions. Its pairings already differ from the original ones by only the allocated error. Exceptional bins contribute at most their probabilities. We have therefore obtained a signed combination of deterministic pinned laws with coefficient mass at most one, uniformly against every bounded far test.

Apply Lemma 3.2 to every member of the finite barrier list. For a joint barrier choice, add the profiles in its disjoint squares. The field-only measure is exactly the original reference measure tilted by the single normalized weight for this sum profile. Normalization is global; no product of local normalizing constants is assumed. The coefficient mass does not change. Allocate the remaining errors among the finitely many choices and squares. This proves (3.5).

All source squares were placed on fixed-factor contours at the observable's end of the original collar. Consequently their profiles remain within Cs at the inner end, or between cR and CR at

the outer end, regardless of how small the deletion disks, strip widths, or approximation tolerances are chosen. They avoid the center. The deletion and subdivision choices put them strictly in the interior, also in boundary charts. This proves the support claims.

If a finite symmetry group preserves the reference law and the geometry, average the replacement over that group, with its appropriate character for an odd observable. Enlarge the finite profile list by the corresponding images. The sum of absolute coefficients cannot increase. In the plane this includes the right-angle lattice rotations and spin flip; free boundary also permits spin flip.

The two-sided finite-array assertion requires uniform integrability when a soft weight replaces one of the original bounded tests. We give its proof after establishing the required moment bounds in Lemma 3.4. The preceding one-sided construction is independent of those bounds. $\square$

3.4 Moments and the order of the two replacements

Lemma 3.4 (Smeared moments and exponential tails). *Let $h_1, \dots, h_m$ be fixed bounded piecewise regular profiles with compact interior supports in a reference chart. All joint polynomial moments of the variables X_{δ, h_i} converge to the integrals of the corresponding continuum spin correlations. For every $t < \infty$,*

$$\sup_{\delta \text{ small}} \mathbb{E} \exp \left(t \sum_{i=1}^m |X_{\delta, h_i}| \right) < \infty. \quad (3.8)$$

Joint exponential expectations converge as well. In particular the weights in (3.3) have convergent normalizations and uniformly integrable powers of every fixed positive order.

For two separated support regions with a fixed buffer, there is a geometric constant K such that

$$\mathbb{E}[AU_{\delta, g}] \leq K \mathbb{E}A \quad (3.9)$$

for every nonnegative function A of the first region and every normalized positive soft weight on the second. The constant is independent of g , its magnitude, and its complexity.

Proof. First consider one compact set. Buffered comparisons reduce bounds on nonnegative functions there to bounds in the plane, or to a fixed homogeneous square reference. The standard two-point estimate is

$$\delta^{-1/4} |\mathbb{E}[\sigma_z \sigma_{z'}]| \leq C(|z - z'| \vee \delta)^{-1/4}$$

in the plane. Its singularity is integrable in two dimensions, so the second moments of the smeared fields are uniformly bounded. The buffered comparison gives the same conclusion in compact interior parts of the other reference geometries, and hence bounded first moments as well.

For a nonnegative profile h , the GHS inequality [23, Theorem 2], in the nonnegative-field form of [7, Theorem 3.6], gives in a homogeneous plus or free reference, for $t \geq 0$,

$$\log \mathbb{E} e^{tX_{\delta, h}} \leq t \mathbb{E} X_{\delta, h} + \frac{1}{2} t^2 \text{Var}(X_{\delta, h}).$$

The coefficients on the right are uniformly bounded. The negative exponential is bounded by the corresponding positive one using spin flip and monotonicity; minus boundary follows by spin flip. Plane and half-plane references follow by taking the homogeneous finite-volume limits. Write a signed profile as $h = h_+ - h_-$ and apply Cauchy–Schwarz to the two exponentials. Covering a compact support by finitely many buffered squares and applying Hölder, if necessary, proves (3.8) for every finite profile list. These are also the smeared-field exponential estimates included in Proposition 2.5; the argument explains why no sign restriction on h is needed.

We next justify moment convergence at diagonals. In a homogeneous plus or free reference, cluster coloring bounds the absolute spin product mean by the probability that every insertion

connects to the boundary of a small square around it. Choose that square's radius to be a fixed fraction of the minimum of its distance to another insertion and its distance to the physical boundary. The squares can be taken disjoint. Conditioning successively outside them and using the fully wired one-arm upper bound gives the product bound

$$\delta^{-b/8} \left| \mathbb{E} \prod_{i=1}^b \sigma_{z_i} \right| \leq C_b \prod_{i=1}^b (d_i \vee \delta)^{-1/8}, \quad (3.10)$$

where d_i is the indicated nearest-point or boundary distance. Repeated lattice sites are allowed by taking the trivial arm at radius δ . Fixed minus boundary is reduced to plus by spin flip.

On compact interior sets, the right side of (3.10) has uniformly integrable collision singularities. To see the relevant power counting, group close points in dyadic clusters. A cluster of $q \geq 2$ points costs at most singular degree $q/8$, while its relative coordinates have dimension $2(q-1)$. At every nontrivial cluster in the nested grouping, $2(q-1) - q/8 > 0$. Summing the resulting geometric series, from the smallest clusters outwards, bounds the integral and makes the contribution of tuples within distance ε of a diagonal tend to zero uniformly in δ . The same counting bounds the lattice sums, with the final dyadic shell truncated at δ .

Away from the diagonals, Proposition 2.5 supplies separated-point convergence, and the regular profile pieces give Riemann-sum convergence. Removing the small diagonal region now proves every joint polynomial moment limit. The exponential bound permits uniform approximation by Taylor polynomials: for example the tail after degree N is bounded in expectation by $2^{-N} \mathbb{E} e^{2|X_{\delta,h}|}$. This proves convergence of exponential expectations and their joint versions. Jensen's inequality and the bounded first moments keep each normalizing denominator bounded away from zero. Higher exponential moments therefore give the asserted uniform integrability of normalized weights.

Finally, the buffered density-ratio bound implies

$$\mathbb{E}[A \mid \text{spins in the second region}] \leq K \mathbb{E} A$$

for every nonnegative A in the first region: compare a fixed second-region assignment to its unconditional mixture. Multiply by the nonnegative normalized second-region weight and integrate. This proves (3.9). Arbitrarily large finite fields are allowed in the pointwise estimate, and the bound does not depend on the chosen profile. $\square$

Completion of the proof of Lemma 3.3. For the finite families in (3.7), first choose the inner replacements with a small error budget. For each row α , the one-sided statement gives one vector $c_{\alpha i}$ that works for all outer columns. The inner profile list is finite and independent of these rows. By Lemma 3.4, choose a finite $Q \geq 1$ such that the tails

$$\mathbb{E}[U_{\delta, h_i} \mathbf{1}_{\{U_{\delta, h_i} > Q\}}]$$

are uniformly small for every profile in those lists and every sufficiently small mesh. Apply the outer one-sided replacement to every $H_{\beta, \delta}$, choosing its error smaller by the factor Q . It gives one vector $d_{\beta j}$ that works simultaneously against all the bounded inner tests $(U_{\delta, h_i} \wedge Q)/Q$. Thus the coefficients factor by row and column, as required in (3.7). The unused fixed-factor buffer margin allows its inner test region to contain the whole support of the first replacement, rather than just the original B_s .

The truncation error against an original bounded outer observable is bounded by the displayed tail. Against any positive normalized outer replacement it is bounded by a geometric constant times that tail, by (3.9). The sum of absolute outer coefficients is at most one. Thus this error estimate

was available *before* the outer profile list was chosen; no bound on the eventual field strengths is needed. Removing the truncations and summing the allocated finite error budgets proves the two-sided assertion, including pairings with the constant one. $\square$

The support statements in Lemma 3.3 leave fixed relative buffers between radial ends, even as the approximation error tends to zero. This feature, together with (3.9), will give uniform Hilbert-space norms in the radial argument. All fine-mesh thresholds here may depend on the fixed scale ratio and the requested error. No quantitative rate of primary-field convergence, necklace convergence, or stopping-band convergence has been used.

4 Radial spectral estimates and uniform conditional averaging

We now turn the finite soft-field replacement into an estimate for conditional averaging of arbitrary bounded lattice observables. All measures in this section are at the nearest-neighbor critical point. Recall that $P_R F$ is the conditional expectation of F given the cutting spins of $B_R(0)$. The bulk estimate uses the response to a fixed-plus disk to detect the leading nonconstant component.

Proposition 4.1 (Uniform bulk conditional averaging). *There are constants $C < \infty$ and $\ell_0 < \infty$ with the following property. Fix $\ell = R/s \geq \ell_0$. For all sufficiently large s , with its threshold allowed to depend on ℓ , let F be any bounded lattice function supported in $B_s(0)$. Put*

$$m(F) = \mathbb{E}_{\mathbb{C}} F, \quad d_R(F) = \mathbb{E}_{\text{disk}(4R)}^+ F - m(F).$$

Then

$$|d_R(F)| \leq C(s/R)^d \|F\|_{\infty}, \tag{4.1}$$

$$\|P_R F - m(F)\|_{\infty} \leq C|d_R(F)| + C(s/R)^p \|F\|_{\infty}, \tag{4.2}$$

with

$$(d, p) = \begin{cases} (1, 3), & F \text{ even and invariant under right-angle rotations,} \\ (1, 2), & F \text{ even,} \\ (1/8, 9/8), & F \text{ odd.} \end{cases}$$

The constants are independent of every sufficiently large fixed ratio. The same conclusion holds along ratios tending to a fixed $\ell \geq \ell_0$, after harmless changes in the thresholds.

The exponent $p = 3$ in the square-symmetric even sector is essential for bulk interactions: summing translated local terms in a larger block costs two powers of the scale ratio. A remainder with exponent two would not give the contraction needed in Section 5. For a marked observable there is no such area factor; the other two estimates have the required strict inequality $p > d$.

We prove the proposition by representing separated pairings in a radial Hilbert space. The spin formulas determine its low spectrum, and a charge cancellation supplies the square-symmetric gain from exponent two to three. Uniform Hilbert norms for the soft weights then make these spectral bounds apply to the finite dictionaries, independently of their complexity. Finally a fixed calibration observable detects the disk component. The half-plane version, proved at the end of the section, gives the corresponding boundary estimate.

4.1 The radial Hilbert space

Write $\mathcal{G}$ for either the plane or the upper half-plane, with homogeneous fixed or free boundary conditions in the latter case. A point state is a finite product

$$S_X = \prod_{x \in X} \sigma(x), \quad X \subset \mathcal{G} \cap \{0 < |z| < 1\},$$

at distinct points. The empty state is denoted by $\mathbf{1}$. We use the continuum spin normalization in the reference limits; an overall per-spin constant will not affect any assertion below. Set $\iota(z) = 1/\bar{z}$. On the linear span of point states define

$$\langle S_Y, S_X \rangle_{\text{rad}} = \left(\prod_{y \in Y} |y|^{-1/4} \right) \mathbb{E}_{\mathcal{G}} \left[\prod_{y \in Y} \sigma(\iota(y)) \prod_{x \in X} \sigma(x) \right], \quad (4.3)$$

with the usual conjugation on the coefficients of the first argument. The primary factors in this definition are those for a field of dimension $1/8$.

The form is positive semidefinite. Indeed, Euclidean reflection positivity for the nearest-neighbor reference model [20] passes from reflection-symmetric critical lattice domains to the separated-point continuum correlations. In the half-plane the reflecting line can be chosen perpendicular to the physical boundary, with both lines lattice aligned before taking the limit. A conformal map conjugates that line reflection to reflection in the unit semicircle and preserves the half-plane. The analogous statement in the plane uses the spherical covariance of the spin formula and a Möbius map taking a line to the circle. Transformation of the field factors gives exactly (4.3). This establishes positivity also for complex linear combinations. Quotienting by the null space and completing will give a Hilbert space $\mathcal{H}$.

For $0 < t \leq 1$, initially on point states, put

$$D_t S_X = t^{|X|/8} S_{tX}. \quad (4.4)$$

Covariance under scaling shows that D_t is symmetric for (4.3), and $D_t D_u = D_{tu}$. We check the contraction property before passing to the quotient. For a finite linear combination v , Cauchy–Schwarz for the positive semidefinite form gives

$$\|D_t v\|_{\text{rad}}^2 = \langle v, D_{t^2} v \rangle_{\text{rad}} \leq \|v\|_{\text{rad}} \|D_{t^2} v\|_{\text{rad}}.$$

For each such v , the quantities $\|D_u v\|_{\text{rad}}$ remain bounded as $u \downarrow 0$. This follows directly from the spin formulas: in the plane each squared product matrix element is a finite sum starting at a nonnegative power of u , as computed in (4.10) below; in the half-plane the height factors cancel the dilation factors and the cross factors have a finite limit, as in (4.12). There is therefore no spectral assumption in this boundedness observation. Iterating the last inequality yields

$$\|D_t v\|_{\text{rad}} \leq \|v\|_{\text{rad}}^{1-2^{-n}} \|D_{t^{2^n}} v\|_{\text{rad}}^{2^{-n}} \longrightarrow \|v\|_{\text{rad}}.$$

If v is null, the first inequality already says that $D_t v$ is null. Consequently D_t descends to a self-adjoint contraction on $\mathcal{H}$. It is positive, since $D_t = D_{\sqrt{t}}^2$. Continuity of separated-point correlations gives strong continuity at $t = 1$ on point states, and then on $\mathcal{H}$ by contraction. Apply the self-adjoint semigroup theorem to $s \mapsto D_{e^{-s}}$, using the underlying real Hilbert space and the complex-linearity of this family; see [6, Theorem 7.3.10 and (7.3.9)]. Contraction makes its generator nonpositive. The spectral calculus [6, Theorem 6.4.1 and Section 6.5] therefore gives a nonnegative self-adjoint operator Δ such that

$$D_t = t^\Delta. \quad (4.5)$$

The vacuum $\mathbf{1}$ has norm one. Spin flip in the plane and in the free half-plane, and rotations in the plane, act isometrically and commute with every D_t .

4.2 A charge identity

We give the algebraic cancellation needed to exclude a rotation-invariant contribution of dimension two. For a list $X = (z_1, \dots, z_n)$ of distinct complex numbers, define

$$w_X(\mu) = \prod_{i < j} |z_i - z_j|^{\mu_i \mu_j / 2}, \quad W_s(X) = \sum_{\substack{\mu_i \in \{-1, 1\} \\ \sum_i \mu_i = s}} w_X(\mu).$$

An impossible charge has weight zero; on the empty list $W_0 = 1$. For an even nonempty list let $\mathbb{E}_0$ mean expectation with probability weights $w_X(\mu)/W_0(X)$ on charge-zero assignments.

Lemma 4.2 (Charge cancellation). *For every even list X ,*

$$\mathbb{E}_0 \left| \sum_i \mu_i z_i \right|^2 = 4 \left(\frac{W_2(X)}{W_0(X)} \right)^2. \quad (4.6)$$

For the empty list both sides are zero.

Proof. We first prove a polynomial identity. On a set I of $2m$ points, write $D_+(\mu)$ and $D_-(\mu)$ for the Vandermonde determinants on the points with the indicated signs, and put

$$V_s(I) = \sum_{\sum_{i \in I} \mu_i = s} |D_+(\mu) D_-(\mu)|^2.$$

Then

$$\sum_{\sum \mu_i = 0} |D_+(\mu) D_-(\mu)|^2 \left| \sum_i \mu_i z_i \right|^2 = 2V_2(I). \quad (4.7)$$

Here and below the Vandermonde on an empty set is one.

For completeness, work in the exterior algebra of $\mathbb{C}^{2m} \otimes \mathbb{C}^2$, with orthonormal site and color basis $e_i \otimes e_{\pm}$. Define

$$p_k^{\pm} = \sum_{i \in I} z_i^k e_i \otimes e_{\pm}, \quad \Psi = \bigwedge_{k=0}^{m-1} (p_k^+ \wedge p_k^-).$$

Let Q project to the subspace with exactly one exterior factor at each site. Both Ψ and $v = Q\Psi$ are invariant under simultaneous $SU(2)$ transformations of the colors: each two-column wedge is multiplied by the determinant, which is one, and Q commutes with color transformations. In the basis indexed by one sign per site, v has only balanced coordinates, whose absolute values are $|D_+(\mu) D_-(\mu)|$.

Let τ_1, τ_2, τ_3 be the Pauli matrices and let $A_j = \sum_i z_i \tau_j^{(i)}$ act on this subspace. The Gram matrix $\langle A_j v, A_k v \rangle$ is invariant under every real three-dimensional rotation, because v is $SU(2)$ invariant. It is therefore a scalar multiple of the identity. For $A_+ = (A_1 + iA_2)/2$ we obtain

$$\|A_3 v\|^2 = 2\|A_+ v\|^2. \quad (4.8)$$

The left side is the left side of (4.7). On the full exterior algebra A_+ is the derivation sending $e_i \otimes e_-$ to $z_i e_i \otimes e_+$ and sending a plus factor to zero; this derivation commutes with Q . Applied to Ψ , all terms vanish by repeated columns except the replacement $p_{m-1}^- \mapsto p_m^+$. The remaining plus columns have degrees $0, \dots, m$, and the remaining minus columns degrees $0, \dots, m-2$. Thus the absolute values of its projected coordinates are exactly the Vandermonde products in $V_2(I)$. Equation (4.8) proves (4.7). When $m = 0$ the identity is the trivial equality $0 = 0$.

We now multiply the numerator on the left of (4.6) by $W_0(X)$. In the resulting sum over two balanced assignments μ, ν , partition the sites into the agreement set $A = \{i : \mu_i = \nu_i\}$ and its complement B . Both restrictions of μ are balanced, since

$$\sum \mu_i = \sum \nu_i = 0 \implies \sum_{i \in A} \mu_i = \sum_{i \in B} \mu_i = 0.$$

The factors joining A to B cancel in $w_X(\mu)w_X(\nu)$. For either group $I = A, B$, put $q(I) = \prod_{i < j \in I} |z_i - z_j|^{-1}$. The remaining factor in that group is

$$\prod_{i < j \in I} |z_i - z_j|^{\mu_i \mu_j} = q(I) |D_+(\mu) D_-(\mu)|^2.$$

If $M_I = \sum_{i \in I} \mu_i z_i$, the mixed terms in $|M_A + M_B|^2$ sum to zero by reversal of all signs on one balanced group. Equation (4.7) therefore shows that the sum under consideration equals

$$2 \sum_{\substack{A \sqcup B = X \\ |A|, |B| \text{ even}}} q(A)q(B) [V_2(A)V_0(B) + V_0(A)V_2(B)]. \quad (4.9)$$

The same grouping applied to two charge-two assignments has charge two on A and charge zero on B . Hence the first sum in brackets is $W_2(X)^2$. Applied to charges two and minus two, it has charge zero on A and charge two on B , so the second sum is $W_2(X)W_{-2}(X)$. Sign reversal gives $W_{-2} = W_2$. Thus (4.9) is $4W_2(X)^2$; division by $W_0(X)^2$ proves the assertion. $\square$

4.3 The low radial spectrum

Proposition 4.3 (Low spectrum). *Let Π_0 be projection onto the vacuum. The following statements hold on the indicated closed sectors of $\mathcal{H}$.*

- (i) *In the even plane sector there is a nonzero rank-one projection Π_1 at dimension one and*

$$\|D_t - \Pi_0 - t\Pi_1\| \leq t^2.$$

On the subspace invariant under rotation by $\pi/2$, the right side may be replaced by t^3 .

- (ii) *In the odd plane sector there is a nonzero rank-one projection $\Pi_{1/8}$ and*

$$\|D_t - t^{1/8}\Pi_{1/8}\| \leq t^{9/8}.$$

- (iii) *In the fixed half-plane, and in the even sector of the free half-plane,*

$$\|D_t - \Pi_0\| \leq t^2.$$

The vacuum matrix element is $\langle v, \Pi_0 w \rangle = \overline{\mathbb{E}v} \mathbb{E}w$.

Proof. We first record how pointwise expansions determine the spectral assertions. For v in the dense span of point states,

$$\langle v, D_t v \rangle = \int_{[0, \infty)} t^q d\nu_v(q), \quad \nu_v \geq 0.$$

Suppose its expansion consists of finitely many specified powers below p , with an $O_v(t^p)$ remainder. The limit at zero identifies the mass at zero. After removing that mass, positivity excludes spectral

mass on each interval below the first remaining specified power: any positive mass on a compact subinterval there would decay more slowly than the expansion allows. Dividing by that first power and taking a limit identifies its atom. Repeating this argument identifies the subsequent atoms and excludes the intervening intervals, including the interval up to p . Polarization identifies their matrix elements. If the excluded spectral projection vanishes on the dense point-state span, it vanishes on the entire Hilbert space. Consequently no uniformity of the constants in the point-state remainders is needed for this step.

Normalize the plane spin field temporarily so that its two-point function is $|z - w|^{-1/4}$. The exact formula for an even list Z is

$$(\mathbb{E}S_Z)^2 = 2^{-|Z|/2}W_0(Z).$$

For lists $X = (x_i)_{i=1}^k$ and $Y = (y_j)_{j=1}^l$ of the same parity, the reflected, dilated matrix element $F_{X,Y}(t) = \langle S_Y, D_t S_X \rangle$ consequently satisfies

$$F_{X,Y}(t)^2 = 2^{-(k+l)/2} \sum_s t^{s^2/4} \sum_{\substack{\sum_i \mu_i = s \\ \sum_j \nu_j = -s}} w_X(\mu) w_Y(\nu) \prod_{i,j} |1 - t x_i \bar{y}_j|^{\mu_i \nu_j / 2}. \quad (4.10)$$

Indeed the inner dilation contributes $t^{(s^2-k)/4}$ to the spin correlation square and $t^{k/4}$ from (4.4). Inversion of the outer list contributes powers of $|y_j|$ that cancel its primary factors and the cross-distance powers. This leaves exactly (4.10). Each remaining cross factor has a convergent integer-power expansion near zero.

For odd lists the lowest charges are $s = \pm 1$. Their leading coefficients are equal and factor as $W_1(X)W_1(Y)$. Taking the positive square root gives

$$F_{X,Y}(t) = a(X)a(Y)t^{1/8} + O_{X,Y}(t^{9/8}),$$

where a is nonzero, since W_1 is positive. In the even sector the charge-zero term has leading coefficient $m(X)^2 m(Y)^2$, where $m(X) = \mathbb{E}S_X > 0$ for a nonempty even list. The coefficient linear in t from charge zero vanishes by sign reversal on one list. The charges $s = \pm 2$ therefore give

$$F_{X,Y}(t) = m(X)m(Y) + m(X)\frac{W_2(X)}{W_0(X)}m(Y)\frac{W_2(Y)}{W_0(Y)}t + O_{X,Y}(t^2). \quad (4.11)$$

The order-one matrix has rank one and is nonzero already for a two-spin state. Empty states are handled directly: their pairings are the means, and they have no nonconstant coefficient.

For the improved even estimate, set $A_X = \sum_i \mu_i x_i$ and $A_Y = \sum_j \nu_j y_j$ under their independent charge-zero weights. The linear cross exponent is

$$-\frac{1}{2}t \operatorname{Re}(A_X \overline{A_Y}).$$

The quadratic term in the logarithm has angular frequency two. Averaging rotation of X over the four right-angle rotations therefore leaves, from the square of the linear term, only

$$\frac{1}{16} \mathbb{E}_0 |A_X|^2 \mathbb{E}_0 |A_Y|^2 t^2.$$

The order- t correction to the $s = \pm 2$ cross factors has angular frequency one, and so supplies no invariant coefficient of order t^2 . Put $a_X = W_2(X)/W_0(X)$ and similarly for Y . Lemma 4.2 says that the invariant expansion of the correlation square, relative to its constant term, is

$$1 + 2a_X a_Y t + (a_X a_Y)^2 t^2 + O(t^3).$$

The order- t coefficient before averaging is already rotation invariant. We may therefore expand the square root first and then average: its invariant quadratic coefficient is

$$\frac{1}{2}(a_X a_Y)^2 - \frac{1}{8}(2a_X a_Y)^2 = 0.$$

Thus the remainder in (4.11) is $O(t^3)$ after projecting one state to right-angle invariance. This proves all the plane spectral statements by the preceding positivity argument. It also identifies Π_0 as the vacuum projection.

In the half-plane, the squares of the fixed/free spin formulas have the form

$$c_n \prod_i (2 \operatorname{Im} z_i)^{-1/4} \sum_{\mu_i = \pm 1} \chi_n(\mu) \prod_{i < j} \left| \frac{z_i - z_j}{z_i - \bar{z}_j} \right|^{\mu_i \mu_j / 2}. \quad (4.12)$$

For fixed plus boundary, $c_n = 2^{-n/2}$ and $\chi_n = 1$. For free boundary and even n , $c_n = (-1)^{n/2} 2^{-n/2}$ and $\chi_n = \prod_i \mu_i$. These are the fixed/free formulas in Proposition 2.5. Fixed minus follows by spin flip.

In a reflected pairing the inner height factors cancel the dilation factor, and the outer height factors cancel the primary factors exactly. At $t = 0$ the cross factors are one. Since $c_{k+l} = c_k c_l$ for all list lengths in the fixed case and for even list lengths in the free case, the constant term is exactly the product of the separate correlation squares.

The linear cross term is odd under simultaneous reversal of the signs on either list. Internal weights are invariant under that reversal; so is $\prod_i \mu_i$ when the list is even. The linear term therefore vanishes. The separate means are nonzero for every fixed-boundary point product, and for every even free-boundary point product. This follows from the positive one- and two-point functions and the ferromagnetic product inequalities, or directly from the formulas. Taking the square root, with the sign prescribed by fixed minus where necessary, gives

$$F_{X,Y}(t) = m(X)m(Y) + O_{X,Y}(t^2).$$

The spectral extraction argument again applies, proving the boundary statement. The zero state and linear combinations are included by direct pairing and polarization. $\square$

4.4 Soft fields as uniformly bounded states

Lemma 4.4 (Uniform radial norms). *Consider the continuum normalized soft weights supplied by Lemma 3.3. If their supports lie on the inner side of a radial circle with a fixed relative buffer, each weight defines an element of $\mathcal{H}$ of norm at most a constant depending only on that buffer geometry. The bound is independent of the bounded profile, its magnitude, and the accuracy used to choose the finite dictionary. A linear combination with total absolute coefficient sum at most B has norm at most CB .*

Proof. For a fixed profile, Lemma 3.4 constructs the exponential by its polynomial series. Integrated spin products belong to the completion of the point-state span: remove small diagonals, approximate the remaining integrals by finite sums of point states, and then restore the diagonals. The integrable product-arm bounds used in Lemma 3.4 control this procedure in the squared Hilbert norm as well, since inner and reflected outer supports are separated. Exponential-series tails converge in that norm by the same moment bounds; absolute values are dominated by a nonnegative majorant profile and the nonnegative fixed-plus or free spin correlations. Fixed minus is reduced to plus by reversing the profile. This proves membership in $\mathcal{H}$, not merely existence of the ordinary expectation.

Let U be such a normalized positive weight and let $\mathcal{J}U$ be its reflected weight, including the Jacobian and primary factors in the reflected profile. Conformal covariance gives $\mathbb{E}\mathcal{J}U = \mathbb{E}U = 1$. On the lattice, the separated-support ratio estimate bounds the conditional mean of either normalized positive weight, given the spins supporting the other, by a fixed constant. Applying this estimate to fixed-profile approximations and taking the limit by Lemma 3.4 gives

$$\|U\|_{\text{rad}}^2 = \mathbb{E}[U\mathcal{J}U] \leq C.$$

Crucially, the lattice bound does not depend on the weights' amplitudes. It therefore holds uniformly over all the finite dictionaries, including dictionaries obtained with smaller approximation errors. The triangle inequality proves the last assertion. $\square$

4.5 Calibration and the lattice operator estimates

We first record the microscopic one-point consequence of the reference-field convergence that will also be used in the renormalization argument. The lattice disk below is centered at the observation site and is invariant under the square symmetries.

Lemma 4.5 (Nonzero disk responses). *Let $\mathcal{O}_{1/8}(z) = \sigma_z$ and let $\mathcal{O}_1(z) = e(z)$ be the symmetric average of the four incident bond products. With m denoting infinite-plane critical expectation, there are nonzero real constants κ_d such that*

$$\lim_{R \rightarrow \infty} R^d \left(\mathbb{E}_{\text{disk}(4R)}^+ \mathcal{O}_d(0) - m(\mathcal{O}_d(0)) \right) = \kappa_d, \quad d \in \{1/8, 1\}. \quad (4.13)$$

Proof. Rescale the disk to radius four and apply the spin and scalar-energy one-point convergence in the reference input Proposition 2.5. Plane centering is the standard energy centering in this one-point statement. The fixed-boundary magnetization at the center is positive, and the plane-centered scalar-energy one-point function there is nonzero. Averaging the four incident orientations preserves its common scalar limit. These observations give (4.13); no rate of convergence is used. $\square$

Proof of Proposition 4.1. Normalize $\|F\|_\infty \leq 1$. For any assignment τ on the cutting contour at radius R , condition first on a separating contour at a fixed smaller radius, for example $R/2$. Relative to the critical plane law, its distribution induced by τ has a density H_τ with

$$\mathbb{E}H_\tau = 1, \quad 0 \leq H_\tau \leq C.$$

The bound follows from the ratio estimate across the intervening fixed-modulus buffer and is uniform over every configuration τ , not just typical configurations. The same construction represents disk-plus expectation by a normalized density H_D on that contour, with the same kind of bound. The Markov property therefore identifies

$$\begin{aligned} P_R F(\tau) - m(F) &= \text{Cov}(F, H_\tau), \\ d_R(F) &= \text{Cov}(F, H_D). \end{aligned} \quad (4.14)$$

Apply the finite-matrix assertion (3.7) to these pairings, scaling the columns by their common sup-norm bound and adjoining a constant row and column to control the separate means. It gives one coefficient vector per row and per column, each with bounded total absolute mass. Lemma 3.4 passes each fixed dictionary to the continuum. The resulting inner and outer states have uniformly bounded norms by Lemma 4.4. Their supports lie at radial ends of radii C_1s and c_1R , where

$C_1, c_1 > 0$ do not depend on replacement accuracy. After rescaling the radial ends, the dilation parameter is

$$t = \frac{C_1 s}{c_1 R}.$$

We choose ℓ_0 large enough that $t < 1$. Parity and, when needed, right-angle invariance are imposed by averaging the inner replacements. These projections commute with D_t , so no symmetry of the outer density is required. Subtracting the separate means removes the vacuum term.

For sufficiently fine mesh, every such finite matrix is therefore, to arbitrarily small absolute error, a matrix of the form

$$t^d a_i b_j + E_{ij}, \quad |a_i|, |b_j| \leq C, \quad |E_{ij}| \leq C t^p, \quad (4.15)$$

by Proposition 4.3. Coefficients in the finite dictionaries may depend on the mesh; along an arbitrary sequence we pass to a subsequence on which they converge. Neither this argument nor (4.15) asserts convergence in Hilbert norm of the original lattice observables. In particular, the first inequality in the proposition follows at once from the disk column.

To obtain the second inequality we show that the disk column has $|b_D|$ bounded away from zero. Adjoin a fixed inner calibration row: in the odd sector take a spin smeared against a nonnegative profile in a small interior box; in the even sector take a product of spins smeared in two small disjoint boxes, and average rotations when appropriate. Normalize these at the inner length scale. They are fixed states of bounded Hilbert norm. Lemma 3.4 and truncation allow them to be paired with the same outer replacements as the bounded row F . Uniform control of the truncation tails against normalized positive outer weights follows from their separated-support density-ratio bound.

Their connected disk responses have a nonzero leading coefficient of order t^d . For the odd calibration this is the positive bulk magnetization in a large fixed-plus disk. For the even calibration, the two-point case of (2.7), with $q = |z - w|/|z - \bar{w}|$, gives the explicit ratio

$$\frac{G_{2,\mathbb{H}}^+(z, w)}{G_{2,\mathbb{C}}(z, w)} = (1 + q)^{1/2} (1 - q^2)^{-1/8} = 1 + \frac{q}{2} + O(q^2).$$

Here $G_{2,\mathbb{C}}(z, w) = |z - w|^{-1/4}$ in the normalization of Proposition 2.5; the ratio is independent of the overall spin normalization. In a disk of radius H centered at zero, the same formula holds with

$$q = q_H(z, w) = \frac{H|z - w|}{|H^2 - z\bar{w}|} = \frac{|z - w|}{H} + O(H^{-3})$$

for points in the two fixed boxes. Indeed a Möbius map to the half-plane gives this q_H , and covariance cancels the primary factors between the disk and plane two-point functions. The coefficient of H^{-1} in their difference is therefore $\frac{1}{2}|z - w|^{3/4} > 0$ in the displayed normalization. Smearing with the fixed positive profiles gives the connected calibration response

$$c_* t^d + o(t^d), \quad c_* > 0, \quad (4.16)$$

as $t \downarrow 0$. This qualitative asymptotic is sufficient for detection; it does not require a lattice convergence rate. Rotation averaging preserves the positive leading coefficient.

Include this calibration row, the row F , the disk column, and the column H_τ in one finite matrix. For all sufficiently small t , (4.16) and (4.15), with replacement error smaller than t^p , give $|a_{\text{cal}} b_D| \geq c > 0$. Since $|a_{\text{cal}}| \leq C$, this implies $|b_D| \geq c/C$. Every other column has $|b_j| \leq C$, whence

$$|t^d a_F b_\tau + E_{F,\tau}| \leq C |t^d a_F b_D + E_{F,D}| + C t^p.$$

By (4.14) this is (4.2), first in the fine-mesh limsup sense.

Finally suppose a claimed lattice bound, with a slightly larger constant, failed at arbitrarily large s for the fixed ratio. Choose the failing functions and, for the supremum estimate, failing contour configurations along that sequence. Apply the finite-matrix argument just given to those rows and columns; the finite dictionaries are uniform over the chosen functions. Choose their accuracy below the fixed remainder scale t^p and then take the fine-mesh limit. This contradicts the limsup bound. It supplies a threshold uniform over F and τ . The same argument applies when the ratios approach the fixed ratio, since all geometric buffers retain their positive clearances. The constants used in the spectral and calibration estimates are uniform for all sufficiently small t , hence for all sufficiently large fixed ratios, as asserted. $\square$

Proposition 4.6 (Uniform boundary conditional averaging). *Fix support and separation factors, including a constant C_0 . There are $C < \infty$ and $\ell_0 < \infty$ such that the following holds. For each fixed $\ell = R/s \geq \ell_0$, there are a lower threshold for s , a sufficiently large chart-window factor, and a sufficiently small flatness threshold, all allowed to depend on ℓ but not on the boundary orientation. Suppose a physical boundary point lies within $C_0 s$ of the common center of the support and cutting squares, and that throughout the chart window the boundary is a single C^1 graph with slope deviation from its tangent below that threshold. Retain its homogeneous nearest-neighbor boundary type in the conditional averaging. Then, for every bounded F supported in the inner square,*

$$\inf_{c \in \mathbb{R}} \|P_R F - c\|_\infty \leq C(s/R)^2 \|F\|_\infty \quad (4.17)$$

in fixed boundary conditions. The same holds for even F in free boundary conditions. For complex F , take $c \in \mathbb{C}$ and adjust the constant harmlessly.

Proof. First take a straight boundary and a fixed homogeneous reference half-plane law. As in (4.14), represent each cutting-contour assignment on a slightly inner contour by a bounded normalized density relative to that law. The physical boundary type is identical in both laws; only the artificial cut is changed. Lemma 3.3 replaces the resulting pairings by uniformly bounded radial states. Subtract the reference mean of F , so that the vacuum projection disappears. The boundary case of Proposition 4.3 bounds every remaining pairing by $C(s/R)^2 \|F\|_\infty$. This is uniform over the arbitrary cut assignment. For fixed minus use spin flip; in the free case use the even projection on the inner state.

The radial origin may be moved to the nearby physical boundary point. The support then lies within a square of radius $(1 + C_0)s$ about that point, while the outer test remains at a distance comparable to R if ℓ_0 is large enough. This changes only the fixed geometric constants. The limiting half-plane may be rotated after taking the continuum limit; covariance makes its radial estimate independent of orientation. The lattice buffers themselves may remain axis-parallel.

For locally graphical curved boundaries use the reference-chart construction in Section 3. Its capped reference domain agrees with the actual physical edge on the buffer under consideration, so the bounded-density representation remains valid. If the asserted thresholds did not exist for a fixed ratio, one could choose a sequence of failing examples with $s \rightarrow \infty$, graph slope deviations tending to zero, and chart windows increasing without bound. After rescaling and passing to a subsequence, the tangent orientations converge and the charts converge locally to a half-plane. The contour replacement and moment convergence statements hold in these charts, with the fixed positive radial buffers described in Lemma 3.3. The same finite-matrix limsup argument as in Proposition 4.1 would then contradict the straight-boundary bound. This proves the existence of a finite window, a positive flatness threshold, and a finite mesh threshold. Compactness of the set of

orientations makes these thresholds uniform in orientation. No rate for convergence of the graph or of a nearest-neighbor correlation is needed. $\square$

Remark 4.7. The order of choices in the two propositions is important. The large ratio is fixed first; approximation tolerances smaller than its spectral remainder are then fixed, and only afterward is the lattice scale required to be sufficiently large. The geometric and spectral constants are uniform over large fixed ratios, but no uniform rate in that last fine-mesh passage has been asserted. For a fixed compact C^2 Jordan boundary, the graph and flatness conditions of Proposition 4.6 hold on all sufficiently small physical windows, uniformly along the boundary.

5 Local changes of scale

All averaging in this section is in the critical nearest-neighbor reference measure. We change the interaction functions on its original spins, preserving the partition integral exactly at each step. The linear part is conditional averaging. The nonlinear corrections form a connected-list expansion, whose bounds are uniform in the volume. The radial estimates will then show that the linear map contracts homogeneous backgrounds after one thermal direction is removed. None of these estimates presupposes correlation bounds in an interacting model.

Fix a large integer L and write

$$r_j = r_0 L^j, \quad r = r_j, \quad R = Lr, \quad M = \lceil (\log L)^2 \rceil.$$

The order of choices is important: L is chosen first, r_0 is then made large enough for the fixed-ratio estimates of the preceding section, and the permissible perturbation is finally made small. Constants denoted by C_L may depend on the fixed value of L but not on j or the volume.

5.1 Carriers, constants, and norms

An interaction is a list $h = (f_{A,z})$. Its carrier A is a finite connected set of lattice sites, its anchor z belongs to A , and $f_{A,z}$ depends only on the spins available in A . In a domain a carrier may extend outside the domain; those exterior sites are not additional variables. On a torus we use connected torus carriers and the torus distance. Terms at an identical index (A, z) are added as functions. We do not require a unique decomposition of a function into different carriers.

For backgrounds, functions are considered modulo constants. We choose their canonical representatives by subtracting their expectation under independent uniform values of the available variable spins. Denote this projection by π . It is linear, preserves the relevant symmetries, and has sup-norm operator bound at most two. The removed scalar is retained whenever sources are present. Thus, in a finite volume, an identity

$$\mathbb{E}_0 e^H = e^c \mathbb{E}_0 e^{H'}$$

includes the accumulated scalar c ; source-dependent constants are never allowed to reenter H' in a later transformation. Background constants cancel upon partition normalization.

For a carrier A , let $A + B_r$ be its max-norm r -neighborhood and put

$$N_r(A) = \frac{|A + B_r|}{r^2}, \quad \|h\|_r = \sup_x \sum_{\substack{A \cap B_r(x) \neq \emptyset \\ z \in A}} \|f_{A,z}\|_\infty e^{N_r(A)}. \quad (5.1)$$

Lattice counting measure is used in N_r . The harmless bounded changes caused by rounding radii are absorbed in constants. These componentwise summation norms give Banach spaces, and the

translation, square-symmetry, and spin-flip restrictions define closed subspaces. The homogeneous background space is translation invariant, invariant under the square lattice symmetries, and spin-flip even.

A source coefficient is supplied with a nonempty set of labels, and its carrier must contain their marked sites. For any one fixed label set its norm is the total-list norm

$$\|u\|_{r,\bullet} = \sum_{A,z} \|u_{A,z}\|_{\infty} e^{N_r(A)}. \quad (5.2)$$

The anchor of a singleton-source list is its marked site. For other marked lists one may choose a mark according to a fixed rule. The background norm has a volume factor when all anchors are summed in a block; the marked norm does not.

We call a carrier short if $N_r(A) \leq M$. A connected carrier satisfies $\text{diam } A \leq CrN_r(A)$: along a path between distant points, select separated r -boxes, each contributing order r^2 to the thickened carrier. Consequently every short carrier anchored at z is contained in $B_{CMr}(z) \subset B_R(z)$ once L is large. All carriers of the distinguished fields below are short.

5.2 Geometry and the exact analytic map

Give an input carrier the protective neighborhood

$$U(A) = A + B_{2R}.$$

Two occurrences in a list are joined when their protective neighborhoods intersect. Connectedness always refers to this overlap graph when it is applied to a list of interactions.

Lemma 5.1 (Carrier geometry). *For connected A and $R = Lr$, with L sufficiently large,*

$$N_R(U(A)) \leq C(1 + L^{-1}N_r(A)). \quad (5.3)$$

Moreover $A + B_{5R}$ can be covered by at most $C_L(1 + N_r(A))$ squares of radius r . These estimates hold on a torus while R is less than a fixed sufficiently small fraction of its side.

Proof. Choose a maximal collection of mutually separated points of A , with separation a fixed multiple of R . Enlarged R -boxes about them cover $U(A) + B_R$, so $N_R(U(A))$ is bounded by a constant times their number. If there is only one such point this gives the constant term in (5.3). Otherwise, from each chosen point a path in A must leave a smaller concentric box. The r -neighborhood of the initial part of that path contains at least cRr sites. Choose these smaller boxes disjoint, including the r -thickening. Summing their contributions to $|A + B_r|$ proves the other term. The argument uses only boxes of size comparable to R and applies unchanged in a torus chart. A coarse r -grid covering of $A + B_{5R}$ costs at most C_L times a covering of $A + B_r$, proving the second assertion. $\square$

The linear rule sends a short input function to $P_{R,z}f_{A,z}$ on carrier $B_R(z)$, and leaves a long input unchanged. At this point constants have not yet been removed. Here and below the cutting contour is included in the square, as in Section 2.1.

Start with finitely many input terms. The connected coefficient estimates below will define the same local rule for arbitrary lists in the rooted-norm ball. To define the nonlinear rule, expand the exponential of the input interaction in ordered lists, with coefficient $1/k!$ for a list of length k . On each singleton component of its overlap graph apply the linear rule. On every component with at least two vertices retain the product of its original input functions. Multiply the component

weights and sum the resulting series, denoted by $\mathcal{W}_j(h)$. This is a function of the reference spins, with constant term one in h . Take its logarithm pointwise, before integrating $\mathcal{W}_j(h)$ against the reference measure.

The coefficient involving two input occurrences illustrates the rule. Write $g_i = P_{R, z_i} f_i$ for a short occurrence and $g_i = f_i$ for a long one. With temporary parameters s, t , the coefficient of st in $\mathcal{W}_j(sf_1 + tf_2)$ is $g_1 g_2$ when the protective neighborhoods are disjoint and $f_1 f_2$ when they intersect. Therefore, before removing constants,

$$[st] \log \mathcal{W}_j(sf_1 + tf_2) = \begin{cases} 0, & U(A_1) \cap U(A_2) = \emptyset, \\ f_1 f_2 - g_1 g_2, & U(A_1) \cap U(A_2) \neq \emptyset. \end{cases} \quad (5.4)$$

Here $[st]$ means extraction of that power-series coefficient. Thus separated occurrences acquire no new interaction, while the correction for an overlapping pair retains its original product. The proof below establishes this connectedness property at every order.

At order at least two assign the connected coefficient to the carrier

$$Q = \bigcup_{i=1}^k U(A_i). \quad (5.5)$$

Average the choice of its output anchor over its input anchors. This preserves translation and square symmetries. For marked coefficients use the chosen marked-anchor convention instead. Unions may be enlarged by a bounded number of lattice sites when needed to make the carrier connected. Finally apply π and record the removed scalars. Write $\mathcal{R}_j$ for the resulting nonscalar map and T_j for its derivative at zero.

For estimates it is useful to retain Q as an auxiliary *test carrier* even at order one: there $Q = U(A)$, although the actual output carrier is $B_R(z)$ or A . This auxiliary set is used only to estimate locality; it does not enlarge the actual carrier of a linearly averaged term.

For source bookkeeping, attach a distinct formal label to each marked insertion and impose $t_\ell^2 = 0$ for each label ℓ . If t_S denotes the product of the labels in S , set $t_{ST} = t_{S \cup T}$ when $S \cap T = \emptyset$, and $t_{ST} = 0$ otherwise. Thus every retained monomial uses each label at most once. The full finite source algebra is specified in Section 7.

Proposition 5.2 (Exact analytic transformation). *For each fixed sufficiently large L , the maps $\mathcal{R}_j$ are analytic on a common complex neighborhood of zero in (5.1), uniformly in j and the finite volume. They satisfy $\mathcal{R}_j(0) = 0$ and*

$$\|\mathcal{R}_j(h) - T_j h\|_R \leq C_L \|h\|_r^2, \quad \|D\mathcal{R}_j(h) - T_j\| \leq C_L \|h\|_r. \quad (5.6)$$

The derivative bound holds also from the marked norm at r to the marked norm at R . For a marked coefficient, the corresponding scalar-increment functional is bounded, and its derivative differs from its derivative at zero by at most $C_L \|h\|_r$ in the same marked norms.

The coefficient estimates remain valid if the output weight $e^{N_R(\text{output carrier})}$ is replaced by $e^{2N_R(Q)}$. In particular, restricting an output list to connected test carriers with diameter at least D gives the additional factor

$$e^{-cD/R}. \quad (5.7)$$

This factor can be retained simultaneously with the norm of any small input, including a marked remainder or a background. The assertion holds before scalar projection as well as after it, and includes long unchanged singletons with their test neighborhoods.

For any fixed number of square-free source labels, the induced maps on the corresponding coefficients are polynomial and uniformly Lipschitz on bounded sets of marked coefficients, while the background remains in this common small ball. All the preceding enhanced-weight and diameter bounds apply to these maps. Constants may depend on the number of labels and their norm bounds.

In finite volume the transformation, including its scalar factor, preserves the partition integral exactly. The same conclusions apply to a preliminary transformation of finite-range microscopic interactions to a fixed large scale r_0 , averaging every singleton there; its constants may also depend on r_0 and the finite microscopic range. Finitely many prescribed source carriers may instead be left unchanged if they are too large for these cutting squares, with constants allowed to depend on those carriers.

Proof. We prove the integral identity first, then bound the connected coefficients and sum them in the weighted norms. The same estimates will also control marked coefficients and the removed scalars.

Integral preservation. If an occurrence is an isolated vertex of a list's overlap graph, its cutting square has a nearest-neighbor buffer from the supports of all other factors. Cutting squares of distinct isolated vertices likewise have such a buffer from one another. Conditional on their cutting contours and the remaining spins, the reference measure makes their interiors independent. Replacing their factors by conditional expectations therefore leaves the list's integral unchanged. A long singleton is not altered. This proves exactness for every list, including lists with nonsingleton components; it does not require the components themselves to be independent in the unconditioned reference measure. Consequently the integral of $\mathcal{W}_j(h)$ is the original partition integral. Exponentiating the scalar and nonscalar parts of $\log \mathcal{W}_j(h)$ recovers $\mathcal{W}_j(h)$ pointwise, giving the required partition identity.

Connected coefficients. Fix k labeled occurrences and their overlap graph G . The transformed moment of any subset of these labels factors over the components of its induced graph. It follows either by the partition formula for a logarithm, or by the following polymer expansion, that the logarithmic coefficient vanishes unless G is connected. For connected G its sup norm is at most

$$2^{k-1} \tau(G) \prod_{i=1}^k \|f_i\|_\infty, \quad (5.8)$$

where $\tau(G)$ is the number of spanning trees of G .

The connected logarithm uses the hard-core Mayer expansion and the Penrose tree-graph bound [18, Section 2 and Proposition 5]. We use the minimum-spanning-tree partition scheme [29, Section 3.2]; the extra factor 2^{k-1} will come from the internal/external edge markings below. To prove (5.8), use square-free variables for the k occurrences. Regard a connected subset of labels as a polymer. Its activity is the product of the original functions if its size is at least two, and the linearly transformed function if its size is one. Polymers are incompatible when their label sets overlap or an edge of G joins them. The hard-core polymer partition polynomial is exactly the transformed moment polynomial, because every induced graph has a unique decomposition into its components.

Its logarithm is a sum over partitions of the labels into connected blocks. For a fixed partition, the absolute Mayer coefficient is bounded by the spanning-tree sum of the quotient contact graph. For completeness, order that graph's edges strictly and group its connected spanning subgraphs according to their greedy minimum spanning tree. For a given tree, the possible extra edges are optional independently: an edge may be added precisely when it is later than every edge on its tree path. Their alternating sum is zero or has absolute value one. Summing over trees proves the bound.

Choose a spanning tree within each connected block, and for every edge of a quotient tree choose an original contact edge. Allowing all these choices only increases the bound. Their union is a spanning tree of G whose edges are marked as internal to a block or between blocks. Conversely, the tree and its edge marks determine the blocks and these choices. There are at most 2^{k-1} markings for each spanning tree of G . Finally each polymer activity is bounded by the product of its input norms, since conditional averaging is a sup-norm contraction. This proves (5.8). Repeated input indices cause no change: their occurrences were labeled, and the original expansion still carries $1/k!$.

Weighted summation. Subadditivity and Lemma 5.1 give

$$2N_R(Q) \leq 2 \sum_i N_R(U(A_i)) \leq Ck + \frac{C}{L} \sum_i N_r(A_i). \quad (5.9)$$

Choose L so large that the last coefficient is, for example, less than one quarter. Thus a fixed fraction of each input exponential weight remains after paying the enhanced output weight.

Root a spanning tree at an input whose test neighborhood meets the output rooted block. Such a root carrier meets $B_{CR}(x)$; covering that square by $O(L^2)$ input rooted blocks costs C_L . If a child touches its parent's protective neighborhood, its carrier meets $A_{\text{parent}} + B_{5R}$. By Lemma 5.1, the sum of possible child terms costs at most $C_L(1 + N_r(A_{\text{parent}}))$ times the child norm. For a vertex with b children, the unused exponential weight pays this factor to the power b , using

$$(1+t)^b e^{-ct} \leq b! C_c^{b+1}, \quad t \geq 0.$$

Reserve fixed fractions of the weight for this inequality and for (5.9); they do not depend on k .

The resulting tree bound is $C_L^k \prod_i b_i!$ times the product of input norms. Summing rooted labeled trees and dividing by $k!$ costs only another constant to the power k : ordering the children identifies these objects with plane rooted trees, whose number on k vertices is the Catalan number $\frac{1}{k} \binom{2k-2}{k-1} \leq 4^{k-1}$. Choices of root and the factor 2^{k-1} are included in C_L^k . Consequently the order- k contribution, including the doubled test weight, is bounded by

$$C_L^k \|h\|_r^k. \quad (5.10)$$

The order-one rule is also bounded with this weight. For its short part the output test neighborhoods have bounded N_R and the root covering cost is C_L ; for its long part use (5.9) directly. Projection costs at most two. Summing (5.10) proves analyticity and (5.6), including the usual Lipschitz bounds on a smaller ball.

If a connected test carrier has diameter at least D , its R -neighborhood has cardinality at least cDR , hence $N_R(Q) \geq cD/R$. The estimate with $e^{2N_R(Q)}$, after retaining the ordinary output weight, leaves (5.7). In the torus the same argument applies to distances up to a fixed fraction of the side; a wrapping or non-chart connected carrier in particular crosses such a distance. For a singleton the same estimate is made on $U(A)$ even when its actual carrier is smaller. Each input norm has remained a separate factor throughout the argument. A small factor in one of them therefore persists when the diameter restriction is imposed.

Marked coefficients and scalars. For marked coefficients, root a tree at an occurrence carrying a specified mark. Summation of this root is by (5.2), and the remaining argument is identical. For multiple labels, regard each input coefficient as carrying the product of its labels and retain only disjoint label sets. There are only finitely many allocations for any fixed total label set. The background occurrences are summed by the convergent series above; the marked occurrences give polynomials in the marked coefficients. Equivalently one can differentiate on a small source polydisc whose radius depends on their norm bound. This proves all the marked analytic and Lipschitz

statements. In particular, terms in a one-mark derivative other than its zero-background linear term contain a background factor, which proves the second bound in (5.6) in marked norm. The removed scalar of a function has absolute value at most its sup norm. Before discarding its carrier, apply exactly the same estimate; this proves the asserted bounds for scalar increments and their background differences. No parity assumption was needed for these marked estimates.

For finitely many input terms the partition identity follows first as an identity of power series near zero. The convergent logarithmic series then defines its exponential throughout the common small ball, and analytic continuation gives the identity there. In a finite system the total absolute norm is finite, bounded by finitely many rooted norms: every nonconstant term meets a variable site. Truncating the list and using the proved convergence gives the identity for all interactions in the ball. This argument does not expand a thermodynamic partition function under the interacting measure.

For the preliminary finite-range transformation, take protective neighborhoods at the fixed radius r_0 and make r_0 larger than every microscopic support. The same exactness and tree proof apply. All covering costs are finite constants depending on this radius and the range; the enhanced output weight costs at most a fixed factor per input occurrence. A prescribed larger source carrier is left unchanged in the singleton rule, with its own protective neighborhood retained. The marked rooted estimates then have finite additional constants depending on that carrier. This proves the last assertion. $\square$

One useful consequence isolates the long linear terms. Let T_j^{long} denote the rule that leaves terms with $N_r(A) > M$ unchanged. Comparing the input and output weights and covering an R -rooted block by r -blocks gives

$$\|T_j^{\text{long}}h\|_R \leq CL^2 e^{-cM} \|h\|_r = o_L(1) \|h\|_r. \quad (5.11)$$

The same conclusion holds for a marked list, also after multiplication by either L or $L^{1/8}$. The constants $c > 0$ and the polynomial covering factor can be chosen independently of sufficiently large L ; the quadratic map constants C_L need not have this property.

5.3 Distinguished local fields

Use the microscopic observables

$$\mathcal{O}_1(z) = e(z) = \frac{1}{4} \sum_{w: |w-z|_1=1} \sigma_z \sigma_w, \quad \mathcal{O}_{1/8}(z) = \sigma_z.$$

For a function in a bulk chart let $m(F)$ denote its plane reference mean, and set

$$d_R(F) = \mathbb{E}_{\text{disk}(4R)}^+ F - m(F).$$

Both quantities here concern the reference model. Scalar changes to F do not change $d_R(F)$.

Lemma 5.3 (Distinguished fields). *On carrier $B_{r_j}(z)$ define*

$$v_j^{(d)}(z) = r_j^d \pi_{P_{r_j, z}} \mathcal{O}_d(z), \quad d \in \{1, 1/8\}. \quad (5.12)$$

For spin the projection removes no constant. If r_0 is sufficiently large, there are constants $0 < c < C < \infty$, independent of j and of the large blocking ratio L , such that

$$c \leq \|v_j^{(d)}(z)\|_\infty \leq C. \quad (5.13)$$

Their single-term marked norms are likewise bounded above and below. For R/r_j sufficiently large,

$$c(r_j/R)^d \leq |d_R(v_j^{(d)}(z))| \leq C(r_j/R)^d, \quad (5.14)$$

where the disk is centered at z .

Proof. Lemma 4.5 gives constants $\kappa_d \neq 0$ such that

$$R^d d_R(\mathcal{O}_d(z)) \longrightarrow \kappa_d. \quad (5.15)$$

Whenever the conditioning square is inside the disk, the Markov property and the tower property give

$$d_R(P_{s,z}\mathcal{O}_d(z)) = d_R(\mathcal{O}_d(z)). \quad (5.16)$$

This proves (5.14) for all sufficiently large radii, and gives the lower bound in (5.13) by taking a disk of radius comparable to r_j . The difference of two expectations of a canonical representative is bounded by twice its sup norm.

For the upper bound, choose a fixed large ratio B , independently of the later blocking ratio L . At all sufficiently large s , apply Proposition 4.1 to $F = P_{s,z}\mathcal{O}_d(z) - m(\mathcal{O}_d)$ and to the outer radius Bs . The relevant remainder exponent is $p = 3$ for the square-symmetric even field and $p = 1 + 1/8$ for the odd field. Equations (5.15)–(5.16) yield

$$(Bs)^d \|P_{Bs,z}\mathcal{O}_d(z) - m(\mathcal{O}_d)\|_\infty \leq C + CB^{d-p}s^d \|P_{s,z}\mathcal{O}_d(z) - m(\mathcal{O}_d)\|_\infty.$$

Take B large enough that the last coefficient is below one. Iterating from radii in a fixed compact interval above the threshold proves a uniform bound for every large radius. Integer radii are handled by ratios tending to B , which are allowed in Proposition 4.1. The initial compact interval costs only a finite constant. Passing from plane centering to π costs at most two. This bound was obtained with B fixed, so is independent of the later choice of L . Finally $N_{r_j}(B_{r_j}(z))$ is bounded above and below by absolute constants, giving the marked-norm assertion. $\square$

Let $\mathbf{v}_j$ be the homogeneous thermal list with one term

$$r_j^{-2}v_j^{(1)}(z) \quad (5.17)$$

at every anchor z . Its background norm is comparable to one: a rooted r_j -block meets order r_j^2 of these carriers. Nested conditional expectations compose, so the linear rule satisfies the exact identities

$$T_j \mathbf{v}_j = L \mathbf{v}_{j+1}, \quad L^d T_j v_j^{(d)}(z) = v_{j+1}^{(d)}(z) \quad \text{in the marked space.} \quad (5.18)$$

All these equalities are modulo constants where appropriate.

5.4 Linear contraction estimates

The background and marked norms count local fields differently. A homogeneous background has order L^2 input blocks per output block, so its contraction requires a radial remainder with dimension greater than two. The right-angle invariant estimate in Proposition 4.1 provides dimension three after the energy response is removed. A marked coefficient has no such volume factor: its prescribed normalization contributes only L for energy or $L^{1/8}$ for spin. The respective remainder dimensions two and $9/8$ therefore suffice. At the physical boundary there are order L blocks along the edge, and the boundary remainder has dimension two. The following proof keeps the additional powers of the short-carrier cutoff M ; since $M = \lceil (\log L)^2 \rceil$, they do not spoil these strict gains.

Proposition 5.4 (Bulk, marked, and boundary linear bounds). *Given $\rho > 0$, choose L sufficiently large and then r_0 sufficiently large. In the homogeneous background space there are uniformly bounded splittings into the line $\mathbb{R}\mathbf{v}_j$ and a complement such that*

$$T_j = \begin{pmatrix} L & B_j \\ 0 & D_j \end{pmatrix}, \quad \|D_j\| \leq \rho, \quad \|B_j\| \leq C_L. \quad (5.19)$$

The coordinates in this splitting are denoted by (t_j, w_j) .

For a singleton marked source, in the even sector with $d = 1$ or the odd sector with $d = 1/8$, the scaled operator $L^d T_j$ has a corresponding splitting about $v_j^{(d)}$ with matrix

$$\begin{pmatrix} 1 & B_j^{(d)} \\ 0 & D_j^{(d)} \end{pmatrix}, \quad \|D_j^{(d)}\| \leq \rho, \quad \|B_j^{(d)}\| \leq C_L. \quad (5.20)$$

The even projections can be chosen invariant under square rotations.

In a finite domain consider a boundary family: every carrier touches a fixed bounded lattice neighborhood of the physical edge. Its ordinary linear image, modulo constants, has norm at most ρ times its input norm, provided the local graph and flatness conditions of Proposition 4.6 hold on the required windows at the current scale. For free boundary conditions the family must be even; for either fixed boundary condition no parity restriction is imposed. The image is again a boundary family.

Proof. We first prove the quotient estimate in the homogeneous plane space. Sum all short input functions at one anchor z into F_z . It is square-symmetric and even, is supported in $B_s(z)$ with $s = CMr$, and obeys

$$\|F_z\|_\infty \leq Cr^{-2}\|h\|_r. \quad (5.21)$$

Indeed, every carrier contains its anchor. For any x , all terms anchored at $z \in B_r(x)$ contribute to the rooted sum in (5.1). Translation invariance therefore makes this sum at least $|B_r|$ times the total weighted sum at a single anchor. This proves (5.21) without a further factor of M .

Subtract $\alpha r^{-2}v_j^{(1)}(z)$ from F_z , choosing α to make its disk response at radius R vanish. Equations (5.14) and Proposition 4.1 show

$$|\alpha| \leq CM\|h\|_r.$$

The adjusted function remains supported in $B_s(z)$ and has norm at most $C(1+M)r^{-2}\|h\|_r$. All its short terms are mapped to the same carrier $B_R(z)$ and are added there. Applying Proposition 4.1 with $(d, p) = (1, 3)$ to this sum, and then projecting off constants, bounds the remaining function by

$$C(CM/L)^3(1+M)r^{-2}\|h\|_r.$$

There are $O(R^2)$ anchors whose output squares meet an R -rooted block, and their output weights are bounded constants. The quotient norm of the short output is consequently at most

$$CL^2(1+M)(CM/L)^3\|h\|_r = o_L(1)\|h\|_r. \quad (5.22)$$

Add the long-term bound (5.11). Since the distinguished line is exactly invariant, the same estimate applies after subtracting any multiple of the input line. Optimizing gives an operator bound $o_L(1)$ on the quotient spaces, rather than merely a bound on a particular representative.

The norm bounds for $\mathbf{v}_j$ give uniformly bounded projections onto these lines: choose a norm-bounded dual functional equal to one on $\mathbf{v}_j$. Choose the functionals first on real lists and extend

them complex-linearly. Their norms remain uniformly bounded, and the resulting projections commute with complex conjugation. The corresponding complement projections have uniform norm. In these splittings the lower-left entry vanishes by (5.18); the quotient estimate bounds the lower-right entry, and the bounded full linear map bounds the upper-right entry. Taking L larger if necessary proves (5.19).

For a singleton-source list sum its short terms at the marked anchor. The sum has norm at most $\|u\|_{r,\bullet}$ and support in B_{CMr} . The same response cancellation subtracts at most $CM^d\|u\|_{r,\bullet}$ times $v_j^{(d)}$. There is now no volume factor. After multiplying by L^d , the quotient bound is

$$CL^d(1+M)^d(CM/L)^p\|u\|_{r,\bullet}, \quad (d,p) = (1,2) \text{ or } (1/8, 1+1/8). \quad (5.23)$$

Both expressions tend to zero with L . The scaled long-term bound also tends to zero. The exact line multiplier is one by (5.18), proving (5.20) as before. Averaging a projection over the finite rotation group preserves its value on the invariant even line and keeps its norm bounded.

Finally let the input be a boundary family. A short carrier touching the edge has its anchor within CMr of it. Proposition 4.6, with support radius $s = CMr$, gives for each such term

$$\|\pi P_{R,z} f_{A,z}\|_\infty \leq C(CM/L)^2 \|f_{A,z}\|_\infty. \quad (5.24)$$

The use of an anchor just off the edge is allowed by that proposition; one can also recenter at a boundary point at distance at most s , changing only fixed support and separation factors.

For a fixed output rooted square, the relevant anchors lie in a band of width CMr along an edge segment of length $O(R)$. In a one-sheet graph chart this band is covered by at most $CL(1+M)^C$ input r -blocks. Each block bounds the sum of the input norms anchored there, since an anchor belongs to its carrier. Summing (5.24) therefore costs at most

$$CL(1+M)^C(CM/L)^2\|h\|_r = o_L(1)\|h\|_r.$$

The long terms are bounded by (5.11). An output square of a short boundary term still touches the boundary, because the anchor is within $CMr < R$ of it; a long unchanged carrier retains its contact. This proves the boundary assertion. $\square$

The three contractions can be arranged simultaneously. The lower threshold on r_0 depends on the chosen L and on the critical buffer estimates. In a fixed bounded domain, the upper permissible physical radius additionally depends on its boundary: a compact C^2 Jordan curve has uniform one-sheet graph charts and arbitrarily small slope deviation on sufficiently small charts. Thus every fixed domain has the windows needed for all sufficiently fine meshes, without a domain-dependent choice of L , r_0 , or ρ .

The boundary contraction in Proposition 5.4 applies to the boundary family itself. Leakage from restricting a bulk family is instead estimated by the bounded map of Proposition 5.2; it is not assigned a spurious contraction factor. This distinction is used in the next section when the critical plane trajectory is compared with finite domains.

6 The critical background and finite-volume comparison

We now choose the temperature by the decay of the homogeneous background. The interpretation of this branch as critical will follow from the correlations; no property of an interacting critical point is assumed in its construction. All maps in this section use the critical nearest-neighbor conditional kernels of Section 2.1.

6.1 A decaying homogeneous trajectory

Choose the contraction parameter in Proposition 5.4 and numbers

$$0 < \rho < \gamma_0 < \gamma < 1. \quad (6.1)$$

Fix the corresponding sufficiently large block ratio L and initial radius r_0 , with r_0 also larger than the range of the fixed potential. The choices may depend on J, V , but not on a domain or on the number of insertions. We may increase r_0 to meet all preceding calibration and buffer thresholds.

Let $\mathcal{H}_j$ be the space of homogeneous, square-symmetric, spin-flip-even background lists at scale r_j . Use the uniformly bounded splitting of Proposition 5.4 to write a background as

$$h = t\mathbf{v}_j + w, \quad w \in W_j.$$

The norm $|t| + \|w\|_{r_j}$ is uniformly equivalent to $\|h\|_{r_j}$. Here t is the coefficient of the effective thermal list; its relation to the microscopic inverse temperature will be determined by the preliminary transformation below. In these coordinates the analytic local map is

$$\begin{aligned} t_{j+1} &= Lt_j + B_j w_j + n_j^1(t_j, w_j), \\ w_{j+1} &= D_j w_j + n_j^2(t_j, w_j), \end{aligned} \quad (6.2)$$

where $\|D_j\| \leq \rho$, $\|B_j\| \leq C_L$, and the remainders are analytic on a common complex neighborhood of zero. They vanish to second order and satisfy uniform quadratic and local Lipschitz bounds there, by Proposition 5.2.

Lemma 6.1 (The decaying graph). *There is $\epsilon > 0$ such that, for every sufficiently small $w_0 \in W_0$, there is exactly one trajectory of (6.2) with this initial stable coordinate, $\sup_j (|t_j| + \|w_j\|_{r_j}) < \epsilon$, and*

$$\sup_{j \geq 0} \gamma_0^{-j} (|t_j| + \|w_j\|_{r_j}) < \infty.$$

It satisfies

$$|t_j| + \|w_j\|_{r_j} \leq C \|w_0\|_{r_0} \gamma_0^j. \quad (6.3)$$

Its initial thermal coordinate $t_0 = \psi(w_0)$ is analytic, with $\psi(0) = 0$. This graph is transverse to the thermal line.

Proof. We give the forward-stable, backward-thermal solution operator, since the spaces and linear maps depend on the scale. Sequence-space constructions for scale-dependent renormalization dynamics are also developed in [2]; the contraction needed here is proved explicitly below. For an imposed forcing $f_j = (f_j^1, f_j^2)$, first solve the stable equation forward:

$$w_j = D_{j-1} \cdots D_0 w_0 + \sum_{i=0}^{j-1} D_{j-1} \cdots D_{i+1} f_i^2,$$

where an empty product is the identity. Then solve the thermal equation backward:

$$t_j = - \sum_{i=j}^{\infty} L^{j-1-i} (B_i w_i + f_i^1). \quad (6.4)$$

On the sequence space with norm

$$\|(t, w)\|_{\gamma_0} = \sup_{j \geq 0} \gamma_0^{-j} (|t_j| + \|w_j\|_{r_j}),$$

these operators are bounded. Indeed the stable forcing sum costs at most $(\gamma_0 - \rho)^{-1} \|f^2\|_{\gamma_0}$, and the backward sum costs at most $(L - \gamma_0)^{-1}(C_L \|w\|_{\gamma_0} + \|f^1\|_{\gamma_0})$. Any solution in this space must obey (6.4): iterate its thermal recurrence to time k and let $k \rightarrow \infty$, using $L^{j-k} t_k \rightarrow 0$.

Substitute $f_j = n_j(t_j, w_j)$ in this bounded linear solution operator. The uniform quadratic bounds imply

$$\|n(t, w)\|_{\gamma_0} \leq C_L \| (t, w) \|_{\gamma_0}^2,$$

and its Lipschitz constant on a sequence ball of radius s is at most $C_L s$. A sufficiently small ball, with radius a fixed multiple of $\|w_0\|_{r_0}$, is therefore invariant and contractive. The contraction theorem proves existence, uniqueness, and (6.3) in this weighted sequence ball. To obtain the stated uniqueness, any two trajectories in the ordinary ϵ -ball with finite weighted norm obey the same linear solution formulas. Their difference is bounded in weighted norm by $C\epsilon$ times itself, by the local Lipschitz bound on the remainders and the bounded solution operator. Choose ϵ so that $C\epsilon < 1$; the constructed trajectory lies in this ball when w_0 is sufficiently small.

The complex analytic contraction, or equivalently the analytic implicit function theorem on this sequence space, gives analytic dependence on w_0 . Evaluation at time zero gives ψ . The tangent space of a graph over W_0 has trivial intersection with the pure thermal line, proving transversality. $\square$

Proposition 6.2 (The temperature branch). *There is $\lambda_0 > 0$ and a real analytic branch $\beta_c : (-\lambda_0, \lambda_0) \rightarrow (0, \infty)$, with $\beta_c(0) = \beta_0$, whose homogeneous effective backgrounds satisfy*

$$h_{j+1} = \mathcal{R}_j(h_j), \quad \|h_j\|_{r_j} \leq C|\lambda|\gamma_0^j. \quad (6.5)$$

The branch depends only on J, V, λ .

Proof. Relative to the nearest-neighbor reference at K_c , the microscopic background in the exponent is

$$(\beta J - K_c) \sum_{\{x,y\}: x \sim y} \sigma_x \sigma_y + \beta \lambda \sum_A V(A) \prod_{x \in A} \sigma_x. \quad (6.6)$$

For the nearest-neighbor temperature displacement, assign half of each unordered bond coefficient to each endpoint, and use its two-site carrier. Thus its anchored function at z is exactly $2(\beta J - K_c)e(z)$. Assign finite connected carriers and anchors to the remaining interaction shapes, distributing coefficients among equivalent anchors when needed. This preserves all lattice symmetries and spin-flip parity. These are lists of coefficients; no infinite-volume partition function is being expanded.

Apply one preliminary version of Proposition 5.2, with protective neighborhoods at radius r_0 and every microscopic singleton averaged on its cutting square B_{r_0} . The finite range makes the same connected-list estimates convergent on a fixed neighborhood of $(\beta_0, 0)$; their constants may depend on r_0, V . This gives an analytic initial list $h_0(\beta, \lambda) \in \mathcal{H}_0$, equal to zero at $(\beta_0, 0)$. The preliminary map is local, has the same reserved exponential weight as the subsequent maps, and preserves the finite-volume integral, with its scalar prefactor retained.

The normalization of its thermal derivative is explicit. Because each unoriented bond occurs twice in the sum of incident bonds,

$$\sum_z e(z) = \frac{1}{2} \sum_{\{x,y\}: x \sim y} \sigma_x \sigma_y.$$

Thus the preliminary derivative with respect to β , at $(\beta_0, 0)$, has the term $2JP_{r_0,z}e(z)$ at every anchor z . By the definition of the homogeneous thermal list in Lemma 5.3,

$$\partial_\beta h_0(\beta_0, 0) = 2Jr_0 \mathbf{v}_0. \quad (6.7)$$

This equality is in the quotient by constants and therefore also holds for the canonical representatives.

Write $h_0(\beta, \lambda) = t_0(\beta, \lambda)v_0 + w_0(\beta, \lambda)$. The scalar analytic equation

$$t_0(\beta, \lambda) - \psi(w_0(\beta, \lambda)) = 0$$

has β -derivative $2Jr_0 \neq 0$ at $(\beta_0, 0)$, since $\partial_\beta w_0(\beta_0, 0) = 0$. The implicit function theorem produces $\beta_c(\lambda)$. All maps commute with complex conjugation, so the branch and trajectory are real for real λ . Shrinking the interval makes β_c positive. Along the branch, $\|w_0\|_{r_0} = O(|\lambda|)$, so Lemma 6.1 gives (6.5). The argument applies to both signs of λ and uses no sign assumption on V . $\square$

6.2 The physical boundary

Let $\Lambda_a = \{z \in \mathbb{Z}^2 : az \in D\}$. Call an ambient carrier A *intact* if $A + B_1 \subset \Lambda_a$, and let $\text{Res}_{a,j} h_j$ retain precisely the terms with intact carriers. All remaining finite-domain terms are represented by connected ambient carriers meeting a fixed-width lattice neighborhood of $\partial\Lambda_a$. A carrier may extend outside Λ_a ; its function depends only on the available variable spins. Terms which are entirely scalar are removed from the background list.

Proposition 6.3 (Boundary backgrounds). *There are constants $C, \lambda_0 > 0$, independent of the allowed domain and of its boundary type, with the following property. For each fixed C^2 Jordan domain D there is $\eta_D > 0$ such that, for sufficiently small a , the effective background with any of the three prescribed boundary conditions has the form*

$$H_j^{a,b} = \text{Res}_{a,j} h_j + b_j^{a,b}, \quad \|b_j^{a,b}\|_{r_j} \leq C|\lambda|\gamma^j, \quad (6.8)$$

through all steps with $ar_j \leq \eta_D$. Every carrier of $b_j^{a,b}$ touches the physical boundary in the above sense. For free boundary conditions its functions are even; for fixed boundary conditions they need not be even.

Proof. A compact C^2 Jordan boundary has uniform one-sheet graph charts whose slope variation is arbitrarily small on sufficiently short physical windows. Choose η_D so that every enlarged local window required by Proposition 5.4 is such a chart up to the stated terminal scale. The normalized chart constants are fixed universally; only the permitted physical radius depends on D . For small a the preliminary scale also lies in these windows.

Before preliminary blocking, the finite-domain prescription differs from the restriction of (6.6) only within the fixed interaction range of the edge. A crossing interaction with fixed exterior spins remains a function of its available spins, with an ambient carrier containing the original interaction set. Deleting crossing interactions gives the corresponding free prescription. Enlarging carriers by a fixed microscopic amount accommodates any symmetry-preserving anchor decomposition. The rooted norm of these discrepancies is $O(|\lambda|)$, because $\beta_c(\lambda) - \beta_0 = O(|\lambda|)$. The local preliminary map therefore gives

$$\|b_0^{a,b}\|_{r_0} \leq C_0|\lambda|. \quad (6.9)$$

The bound uses local density of interaction shapes and the fixed preliminary scale, not the total length of the boundary.

We verify the locality needed at subsequent steps. For an intact short-singleton output, its cutting square and all nearest-neighbor edges required for conditional averaging are present in Λ_a . The conditional kernel is exactly the plane kernel at that square. For an intact connected output of order at least two, its carrier contains all the protective neighborhoods of its inputs, and hence

every reference neighborhood used in its coefficient. All these inputs and kernels also agree with their plane counterparts. A long unchanged singleton uses no outside kernel. Thus coefficients agree on intact output carriers. Any omission or discrepancy in comparing the two maps has a carrier touching the edge; the same is true of an output involving a boundary-family input.

The canonical representative subtracts the independent uniform-spin mean on the available variables. On an intact carrier that set of variables is identical in the plane and the domain, so this subtraction also agrees exactly. At the edge the discarded scalar keeps its generation carrier for any source bookkeeping. Subtracting constants therefore neither produces an interior discrepancy nor transports an edge discrepancy into an interior carrier.

Let $\mathcal{R}_j^{a,b}$ denote the finite-domain map. The preceding locality and the absolute coefficient estimates of Proposition 5.2 give the bulk leakage bound

$$\left\| \mathcal{R}_j^{a,b}(\text{Res}_{a,j} h_j) - \text{Res}_{a,j+1} \mathcal{R}_j(h_j) \right\|_{r_{j+1}} \leq C_L \|h_j\|_{r_j}. \quad (6.10)$$

No boundary contraction is asserted for this difference. In contrast, the linear image of the boundary family itself has norm at most $\rho \|b_j^{a,b}\|_{r_j}$ by Proposition 5.4. The nonlinear terms involving that family cost at most $C_L(\|h_j\|_{r_j} + \|b_j^{a,b}\|_{r_j}) \|b_j^{a,b}\|_{r_j}$. Consequently, writing $q_j = \|b_j^{a,b}\|_{r_j}$, we have

$$q_{j+1} \leq \rho q_j + C_L \|h_j\|_{r_j} + C_L(\|h_j\|_{r_j} + q_j) q_j. \quad (6.11)$$

Evenness in the free case is preserved by every operation. In the fixed cases the unrestricted boundary estimate in Proposition 5.4 applies instead.

Choose α with $\rho < \alpha < \gamma$. On a sufficiently small common ball the last term in (6.11) is at most $(\alpha - \rho)q_j$. Choose $B \geq C_0$ large enough that $C_L C \leq B(\gamma - \alpha)$, where C is the constant in (6.5). Since $\gamma_0^j \leq \gamma^j$, induction then gives $q_j \leq B|\lambda|\gamma^j$. Decrease λ_0 so this bound and (6.5) stay in the analytic balls and in the preceding smallness condition. Every constant in this choice is independent of D ; the domain affects only η_D and the threshold for the mesh. This closes the induction and proves (6.8). $\square$

6.3 Periodic projection and exponentially small discrepancies

Let $\mathbb{T}_N = (\mathbb{Z}/N\mathbb{Z})^2$. Fix a small numerical $\kappa > 0$, for example $\kappa = 1/100$. For a homogeneous plane list, define Π_N by retaining the carriers of diameter less than κN and projecting their sites to $\mathbb{T}_N$. Use one representative of each anchor modulo $N\mathbb{Z}^2$, retaining its relative carrier shapes and their multiplicities. This projects a homogeneous list once; it does not sum the already homogeneous list over all its translates again. Connected retained carriers lift uniquely after their anchor lift is specified. For $r \leq \vartheta N$ with ϑ sufficiently small, comparison in a bounded number of isometric charts gives

$$\|\Pi_N h\|_r \leq C \|h\|_r. \quad (6.12)$$

Proposition 6.4 (Periodic backgrounds). *For a sufficiently small fixed $\vartheta > 0$, the periodic effective background H_j^N satisfies*

$$\|H_j^N\|_{r_j} \leq C|\lambda|\gamma^j, \quad (6.13)$$

$$\|H_j^N - \Pi_N h_j\|_{r_j} \leq C|\lambda|\gamma^j \exp(-cN/r_j), \quad (6.14)$$

whenever $r_j \leq \vartheta N$. The constants and the coupling interval are independent of N, j . On every fixed finite set of scales, coefficients in a fixed plane chart converge to their plane counterparts as $N \rightarrow \infty$, with the tails interpreted in the weighted list norms.

For real λ these backgrounds are real. At every finite volume the successive maps, including the preliminary map, retain the exact partition identity with their scalar prefactors, and the residual tilt defines a positive Gibbs probability.

Proof. Choose ϑ sufficiently small, depending on the already fixed map parameters, that every retained carrier with its norm neighborhood and every short-term cutting window under consideration lies in an isometric torus chart. The periodic microscopic interactions are their usual images; only N larger than the interaction range and the preliminary scale is relevant here.

Compare one periodic map applied to $\Pi_N h_j$ with $\Pi_N \mathcal{R}_j(h_j)$. A connected list for which the entire union of protective and test neighborhoods lies in a sufficiently small max-ball lifts uniquely, up to an overall period. Its input coefficients and contacts agree with those in the plane. The local conditional kernels on all singleton cutting squares agree as well, and canonical uniform-spin subtraction uses identical variables. The resulting coefficients therefore agree whenever neither an input nor the output is removed by the diameter truncation. This argument uses equality of conditional kernels in a chart, not equality of plane and torus marginal laws.

Every exception has extent at least a fixed multiple of N : it contains an omitted input or output carrier, or its connected union of test neighborhoods leaves every such small chart, including by winding around the torus. In the plane or torus where that list is being estimated, the reserved exponential weight in Proposition 5.2 therefore supplies $\exp(-cN/r_{j+1})$. Rooting near a norm block costs only the usual local covering constant. A long unchanged singleton is compared directly with its truncated counterpart and has the same tail bound. Summing all exceptional lists, using the small analytic ball to sum higher powers of the background, gives

$$\left\| \mathcal{R}_j^N(\Pi_N h_j) - \Pi_N \mathcal{R}_j(h_j) \right\|_{r_{j+1}} \leq C_L \|h_j\|_{r_j} \exp(-cN/r_{j+1}). \quad (6.15)$$

The retained small carriers and their neighborhoods have comparable norm weights in the two spaces by the chart choice. The preliminary finite-range map satisfies the same comparison, giving

$$\left\| H_0^N - \Pi_N h_0 \right\|_{r_0} \leq C_0 |\lambda| \exp(-cN/r_0). \quad (6.16)$$

Here is the summation that preserves a spatial exponential after propagation. Set $E_j = \left\| H_j^N - \Pi_N h_j \right\|_{r_j}$ and enlarge the uniform local Lipschitz constant to $K \geq 1$. As long as the compared backgrounds remain in the analytic balls, (6.15) and (6.5) imply

$$E_{j+1} \leq K E_j + C_1 |\lambda| \gamma^j \exp(-cN/r_{j+1}).$$

For a terminal index j , put $x = N/r_j \geq \vartheta^{-1}$ and $q = K/\gamma$. Iteration, including (6.16), gives

$$\frac{E_j}{|\lambda| \gamma^j} \leq C_0 q^j e^{-cxL^j} + \frac{C_1}{\gamma} \sum_{m=0}^{j-1} q^m e^{-cxL^m}. \quad (6.17)$$

At $\lambda = 0$ the discrepancy is identically zero, so the displayed division is needed only when $\lambda \neq 0$. For $0 < c' < c$,

$$\sum_{m \geq 0} q^m e^{-cxL^m} \leq e^{-c'x} \sum_{m \geq 0} q^m \exp(-(cL^m - c')/\vartheta) \leq C_{L,K,\gamma,\vartheta} e^{-c'x}.$$

The series converges because its exponential decay in L^m dominates q^m . The initial term in (6.17) is its $m = j$ term and has the same bound. This proves (6.14), after renaming c' . Together with (6.12) it gives (6.13). Decreasing the common λ_0 makes these bounds stay strictly inside all analytic balls, and closes the finite-scale induction uniformly in N .

For fixed j , (6.14) tends to zero as $N \rightarrow \infty$. The omitted plane carriers also have vanishing weighted tails at that fixed scale. Identifying the remaining local carriers with their unique lifts proves the claimed coefficient convergence. Finally the preliminary and subsequent maps preserve real coefficients and the exact finite-volume identity of Proposition 5.2. Exponentiating the real residual background gives a strictly positive weight, completing the last assertion. $\square$

7 Sources, stopping scales, and correlation limits

The background flow has selected a temperature, but has not yet identified any interacting correlation. We now insert finitely many sources into the exact partition-function identity. Locality is important twice: it makes each one-source normalization independent of the domain, and it prevents distinct ideal sources from interacting before a suitably chosen stopping scale. The remaining critical reference measure retains all of their long-distance correlations.

7.1 Finite source jets and scalar prefactors

Let the source labels be $\{1, \dots, q\}$, where $q \geq 1$ is fixed. The microscopic observable X_ℓ is either a spin at its marked site or one of the four incident bond products, and its dimension is respectively $d_\ell = 1/8$ or $d_\ell = 1$. Write

$$t_S = \prod_{\ell \in S} t_\ell, \quad d(S) = \sum_{\ell \in S} d_\ell \quad (\emptyset \neq S \subseteq \{1, \dots, q\}).$$

We calculate in the finite algebra

$$\mathcal{A}_q = \mathbb{C}[t_1, \dots, t_q] / (t_1^2, \dots, t_q^2).$$

Thus coefficients are ordinary mixed derivatives at zero, with each label used at most once. Products with overlapping label sets vanish. Exponentials and logarithms with zero and unit constant term, respectively, are finite polynomials in this algebra. In particular, no convergence assertion in the source amplitudes is hidden in the use of a finite jet.

In a finite system F , let H_j^F be its nonscalar background at scale r_j , and let ν_j^F be the critical nearest-neighbor reference measure tilted by its total representative. A source coefficient U_j^S is a marked list, with every carrier containing all the marks in S . We use the same notation for the absolutely convergent sum of its functions. Scalar coefficients are denoted by c_j^S . The exact identity at scale j is

$$\begin{aligned} G_j^F(t) &:= \mathbb{E}_{\lambda, F} \exp \left(\sum_{\ell=1}^q t_\ell r_j^{d_\ell} X_\ell \right) \\ &= \exp \left(\sum_{\emptyset \neq S} c_j^S t_S \right) \mathbb{E}_{\nu_j^F} \exp \left(\sum_{\emptyset \neq S} U_j^S t_S \right). \end{aligned} \tag{7.1}$$

Here the expectation on the first line is at the critical branch from Proposition 6.2, with the finite-system prescription under consideration. The preliminary transformation establishes this identity at $j = 0$.

For precision, apply the local map to $H_j^F + \sum_S U_j^S t_S$. If its newly produced nonscalar and scalar source coefficients are $\hat{U}_j^S$ and a_j^S , then

$$U_{j+1}^S = L^{d(S)} \hat{U}_j^S, \quad c_{j+1}^S = L^{d(S)} (c_j^S + a_j^S). \quad (7.2)$$

These factors express the substitution $t_\ell \mapsto L^{d_\ell} t_\ell$ in G_j^F . They are compatible with products because $d(S)$ is additive on disjoint label sets. The scalar prefactor is removed before the next local map is applied. It is therefore only rescaled and added to; it never becomes an input to a nonlinear coefficient. Scalar background contributions cancel between numerator and denominator in the normalized expectation.

The marked conclusions of Proposition 5.2 apply to these operations. For a fixed q , the map on the source coefficients is polynomial, with bounded derivatives on bounded sets of marked norms, uniformly in the scale as long as the background stays in its fixed small ball. This follows either by decorating the convergent list expansion with the labels, or by Cauchy estimates on a sufficiently small source polydisc at that background. The polydisc and the resulting constants may depend on q and the source bounds; the background ball does not. A one-source nonscalar map and its scalar-increment functional differ from their zero-background versions by at most $C_L \|H_j^F\|_{r_j}$ in operator norm. Scalar increments retain their generating carriers for estimates before those carriers are discarded. In particular, their sums have no volume factor: any one of their marks roots the entire list.

Let $\mathcal{E}$ be the set of energy labels. On logarithmic jets define

$$\mathcal{Q}K = K - \sum_{\ell \in \mathcal{E}} t_\ell [t_\ell] K.$$

Then the exact centering identity is

$$[t_1 \cdots t_q] \exp(\mathcal{Q} \log G_j^F) = r_j^{d(\{1, \dots, q\})} \mathbb{E}_{\lambda, F} \left[\prod_{\ell \notin \mathcal{E}} X_\ell \prod_{\ell \in \mathcal{E}} (X_\ell - \mathbb{E}_{\lambda, F} X_\ell) \right]. \quad (7.3)$$

Indeed, replacing an energy by its centered version subtracts precisely $t_\ell r_j^{d_\ell} \mathbb{E}_{\lambda, F} X_\ell$ from the logarithm of its generating function. By (7.1), this coefficient is

$$[t_\ell] \log G_j^F = c_j^{\{\ell\}} + \mathbb{E}_{\nu_j^F} U_j^{\{\ell\}} = r_j^{d_\ell} \mathbb{E}_{\lambda, F} X_\ell.$$

Thus centering removes both the extracted scalar and the residual one-source mean. The scalar coefficients $c_j^{\{\ell\}}$ for energy labels may be omitted throughout the estimates; applying $\mathcal{Q}$ to the remaining logarithm still removes the residual mean. All spin singleton scalars, and all multiple-label scalars, must still be kept. The actual domain and boundary-state means in (7.3) are not replaced by plane means.

7.2 The common one-source normalizations

Let u_j^ℓ be the nonscalar plane singleton list for the microscopic source X_ℓ on the critical background trajectory h_j . Start with the degree-one coefficient of the same preliminary transformation as in (7.1), normalized by $r_0^{d_\ell}$, and set

$$u_{j+1}^\ell = L^{d_\ell} D\mathcal{R}_j(h_j) u_j^\ell.$$

Scalar terms are recorded separately as above; energy lists are taken modulo constants. This definition uses the local list map, before any interacting thermodynamic state is constructed.

Proposition 7.1 (One-source limits). *There are real analytic functions $Z_\sigma(\lambda)$ and $Z_\epsilon(\lambda)$ on a neighborhood of zero, with*

$$Z_\sigma(0) = Z_\epsilon(0) = 1,$$

such that the plane one-source lists on the critical trajectory satisfy

$$u_j^\ell = Z_\ell(\lambda)v_j^{(d_\ell)} + \eta_j^\ell, \quad \|\eta_j^\ell\|_{r_j, \bullet} \rightarrow 0. \quad (7.4)$$

Here Z_ℓ is Z_σ for a spin and Z_ϵ for any incident bond orientation; energy lists are understood modulo constants. The convergence is locally uniform in the coupling. The two functions are nonzero on one smaller coupling interval, independently of domains, boundary conditions, and the number of sources.

Proof. Use the bounded splittings from Proposition 5.4, with the distinguished vectors from Lemma 5.3. For a normalized one-source coefficient the zero-background matrix is

$$A_j = \begin{pmatrix} 1 & B_j \\ 0 & D_j \end{pmatrix}, \quad \|B_j\| \leq C_L, \quad \|D_j\| \leq \rho < 1.$$

The derivative on the critical plane trajectory differs from A_j by an operator E_j with

$$\|E_j\| \leq C_L \|h_j\|_{r_j} \leq C|\lambda|\gamma^j.$$

Products of consecutive A_j are uniformly bounded: their lower-right blocks contract geometrically and their upper-right blocks are bounded by $C_L \sum_{s \geq 0} \rho^s$. Expanding a perturbed product by the locations of its factors E_j therefore bounds it by a constant times $\prod_j (1 + C\|E_j\|)$. This product is finite, uniformly for the coupling in a sufficiently small complex disc. Thus the one-source trajectory is bounded.

Write its coordinates as α_j, w_j . The stable recurrence gives

$$\|w_j\| \leq \rho^j \|w_0\| + C|\lambda| \sum_{i < j} \rho^{j-1-i} \gamma^i.$$

This tends to zero and is summable in j ; it is bounded by $C\theta^j$ for any fixed $\max(\rho, \gamma) < \theta < 1$, after enlarging C . Moreover

$$|\alpha_{j+1} - \alpha_j| \leq C_L \|w_j\| + C|\lambda|\gamma^j.$$

Hence α_j converges, and subtracting its limit times $v_j^{(d_\ell)}$ proves (7.4). The preliminary source map and every later derivative are analytic in the coupling. The bounds just given are uniform on a smaller complex disc, so the limiting scalar is analytic there. Reality of all maps on real inputs gives real values on the real axis. Spin-flip symmetry makes every plane spin singleton scalar contribution identically zero.

Translation covariance makes the limit independent of the marked site. Choose the energy splitting equivariantly under the square symmetries. Rotation about the mark carries the four incident bond sources into one another and fixes the scalar energy line, so their limit scalars agree. At zero coupling the spin source is exactly on its distinguished line after the preliminary step. The average of the four bond sources is $e(z)$, so this average likewise starts exactly on the energy line. The normalized line multiplier is one. Linearity of a one-source derivative and equality of the four scalars give both asserted intercepts. Continuity now gives simultaneous nonvanishing after one decrease of the coupling interval. All choices were made on the plane background and for the two field types, without reference to a domain or a collection of other marks. $\square$

7.3 Separated sources before they interact

We next consider the sites associated with a fixed configuration of distinct macroscopic interior points. Let Δ_* be a lattice-unit separation no larger than any pairwise mark distance or any mark-to-edge distance. In the plane comparison on a torus, it is also much smaller than the side N . It can be chosen comparable to a^{-1} for each fixed configuration. Bounded rounding errors at the marks are harmless after decreasing this separation by a fixed amount.

Choose the last scale r_k below a sufficiently small fixed multiple of Δ_* and below the physical boundary-chart thresholds in Proposition 6.3. On a torus also require r_k below the chart threshold in Proposition 6.4. For a fixed domain and configuration, and in the torus regime $N/\Delta_* \rightarrow \infty$, these choices give

$$k \rightarrow \infty, \quad 0 < c_1 \leq ar_k \leq c_2 < \infty. \quad (7.5)$$

The constants may depend on the fixed configuration and domain. The small stopping fraction is chosen so that all ideal source supports and their protective neighborhoods are disjoint and interior at every step ending no later than k .

Proposition 7.2 (Separated-source comparison). *At the stopping scale just described, the exact finite-system source coefficients satisfy*

$$U_k^{\{\ell\}} = Z_\ell v_k^{(d_\ell)} + o(1) \quad \text{in marked norm,} \quad (7.6)$$

$$\|U_k^S\|_{r_k, \bullet} = o(1) \quad (|S| \geq 2), \quad (7.7)$$

$$c_k^S = o(1) \quad (|S| \geq 2 \text{ or } S \text{ is a spin singleton}). \quad (7.8)$$

The ideal functions in (7.6) are restricted to the system, with their cutting squares contained in its interior. On tori the conclusions hold as $a \rightarrow 0$ and $N/\Delta_* \rightarrow \infty$. All errors use the running normalization $r_k^{d(S)}$.

Proof. Let Π_F denote restriction to intact carriers in a domain, or the small-diameter periodic projection used in Proposition 6.4. At each scale use the trial coefficients

$$W_j^{\{\ell\}} = \Pi_F u_j^\ell, \quad W_j^S = 0 \quad (|S| \geq 2), \quad \tilde{c}_j^S = 0$$

for the scalar labels retained in (7.8). Let $\mathcal{I}$ be this collection of scalar label sets: all sets with at least two labels, and every spin singleton. On source states use the finite product norm

$$\|(U, c)\|_{j, \text{src}} := \sum_{\emptyset \neq S} \|U^S\|_{r_j, \bullet} + \sum_{S \in \mathcal{I}} |c^S|.$$

Write $U_j = (U_j, c_j)$ for the exact state and $W_j = (W_j, \tilde{c}_j)$ for the trial state. Let Φ_j^F be the source step (7.2) with the actual finite-system background H_j^F held fixed. Thus no background error is included among these source variables. Define the error and the one-step trial defect by

$$\mathfrak{e}_j = U_j - W_j, \quad \mathfrak{d}_j = \Phi_j^F(W_j) - W_{j+1}.$$

Energy singleton scalars are omitted from these states, consistently with (7.3).

The defect is bounded by

$$\|\mathfrak{d}_j\|_{j+1, \text{src}} \leq \varepsilon_j \exp(-c\Delta_*/r_{j+1}), \quad \sup_j \varepsilon_j < \infty, \quad \varepsilon_j \rightarrow 0. \quad (7.9)$$

We verify both factors; a fixed small stopping fraction alone would not make the exponential in (7.9) tend to zero.

First, the list/log map has connected generating carriers. A coefficient joining two distinct marks must span their separation. A discrepancy between an interior singleton calculation and its plane calculation must reach the physical edge; a term containing a boundary-family input likewise connects its mark to the edge. In the torus comparison a projection exception must have a nonlocal or winding generating neighborhood. These are precisely diameter conditions covered by the doubled-weight and marked estimates in Proposition 5.2. They give the exponential factor in (7.9), including before scalar projection. They also cover long unchanged singleton carriers.

For the vanishing factor, terms containing a background input have the extra bound $C|\lambda|\gamma^j$, by Propositions 6.2 and 6.3. The corresponding torus bounds follow from Proposition 6.4; in particular its background discrepancy has the stronger norm bound

$$C|\lambda|\gamma^j \exp(-c'N/r_j),$$

which supplies a spatial gain even when that discrepancy is used as one input without reestablishing its generating history.

For terms with no background input, decompose every trial singleton by Proposition 7.1 into its ideal term and remainder. A term containing at least one remainder has a factor tending to zero in marked norm. This factor is retained simultaneously with the spatial exponential: the latter is paid from the reserved carrier weights, whereas the former is one of the multilinear input norms. A term containing only ideal singletons cannot be exceptional or join distinct marks. Each such singleton and its protective neighborhood has radius $O(r_{j+1})$ about its own mark. The stopping fraction makes these neighborhoods pairwise disjoint and interior, so the logarithm of their factorizing list weight has no multiple-label coefficient. Their singleton transfers are exactly the plane ones.

There is no missing spin scalar in this comparison. In the plane it is zero by parity. On intact interior carriers the local graph and the available-spin representative projection are the same in the finite system and on the plane. An interior ideal spin therefore produces no scalar discrepancy. Every remaining spin singleton discrepancy reaches the edge or involves the torus discrepancy or a remainder, and is covered by the preceding estimates. Multiple-label scalar coefficients have the same connected generating carriers as their nonscalar counterparts, so the same proof applies to them. Energy singleton scalars are deliberately absent from the error state by (7.3). For example, after decreasing c one may take a bounded multiple of $|\lambda|\gamma^j + \sum_\ell \|\eta_j^\ell\|_{r_j, \bullet}$ as ε_j . This proves (7.9).

The preliminary defect obeys $C \exp(-c\Delta_*/r_0)$, with a fixed prefactor, by the preliminary local-list estimates. The microscopic sources are interior and separated. On a unit neighborhood of the bounded trial jets, the source maps have a scale-independent Lipschitz constant $K \geq 1$. It includes all the finite multipliers $L^{d(S)}$ for the retained scalars. Consequently $\|\mathbf{e}_l\|_{l, \text{src}}$ at any $l \leq k$, as long as the state stays in this neighborhood, is bounded by

$$CK^l e^{-c\Delta_*/r_0} + C \sum_{i < l} K^{l-i-1} \varepsilon_i e^{-c\Delta_*/r_{i+1}}. \quad (7.10)$$

Put $q_* = \inf \Delta_*/r_k > 0$. At $l = k$, reindexing the sum by $h = k - i - 1$ bounds it by

$$C \sum_{h=0}^{k-1} K^h \varepsilon_{k-h-1} e^{-cq_* L^h}.$$

The majorant $C \sup_i \varepsilon_i K^h e^{-cq_* L^h}$ is summable. For each fixed h the corresponding ε_{k-h-1} tends to zero, so dominated convergence gives an $o(1)$ bound. The same bound is uniform over $l \leq k$: for l

large use the same tail majorant and the smallness of the late ε_i , while the finitely many early l have $\Delta_*/r_l \rightarrow \infty$. The preliminary term also tends to zero uniformly, since geometric growth of Δ_*/r_0 dominates K^k . Thus the estimates bootstrap inside the unit neighborhood. Together with (7.4), they prove all three conclusions. No rate of convergence for that remainder was needed. $\square$

7.4 Evaluation in a bounded domain

At the terminal scale, the entire nonscalar background in a fixed bounded domain has sup norm tending to zero. To see this directly, cover the available lattice sites by $O(1)$ squares of radius r_k ; the number is bounded by (7.5). Every surviving nonconstant representative depends on an available spin, hence its carrier meets one of these squares. Therefore

$$\left\| \sum_A H_{k,A}^F \right\|_\infty \leq C_F \left\| H_k^F \right\|_{r_k} \leq C_F |\lambda| \gamma^k \rightarrow 0. \quad (7.11)$$

The sum here includes both restricted bulk and boundary lists. The covering argument remains valid for carriers extending outside the domain, because a nonconstant term must still contain an available variable spin.

All terminal backgrounds are real for real λ . Their tilts are thus positive probability measures. If a real exponent has sup norm at most b , its normalized density lies between e^{-2b} and e^{2b} . Equation (7.11) consequently permits replacing ν_k^F by the ordinary critical reference measure in expectations of uniformly bounded terminal tests, with error $o(1)$.

Taking the logarithm of the last factor in (7.1) gives a finite polynomial in moments of the source coefficients, through the degrees in use. Their marked norms bound the sums of sup norms. Proposition 7.2 and the preceding probability comparison therefore give, coefficientwise,

$$\mathcal{Q} \log G_k^F = \mathcal{Q} \log \mathbb{E}_{0,F} \exp \left(\sum_{\ell=1}^q t_\ell Z_\ell v_k^{(d_\ell)} \right) + o(1). \quad (7.12)$$

Here and below coefficientwise $o(1)$ refers to the finitely many coefficients of positive degree. The scalar conclusions of Proposition 7.2 are essential in this identity; energy singleton scalars disappear under $\mathcal{Q}$, while spin singleton scalars do not disappear by convention.

The cutting squares of the ideal fields are disjoint and interior. Condition on their exterior, retaining their cutting contours. The critical nearest-neighbor Markov property makes their interiors conditionally independent. For every subset of labels, insertion of the conditional expectations $P_{r_k, z_\ell} \mathcal{O}_{d_\ell}(z_\ell)$ is therefore exactly equivalent to insertion of the microscopic $\mathcal{O}_{d_\ell}(z_\ell)$. This is an equality of all the subset moments, not only of the full product. The representatives used for energy may differ by constants, which leave $\mathcal{Q} \log$ unchanged. Exponentiating (7.12) and using (7.3) identifies the requested centered product, at running normalization, with the corresponding critical product of spins and symmetric incident-bond averages, multiplied by $\prod_\ell Z_\ell$, up to $o(1)$.

Passing to the normalization a^{-d_ℓ} multiplies the error by $(ar_k)^{-d(\{1, \dots, q\})}$, which is bounded by (7.5). The nearest-neighbor convergence in Proposition 2.5 applies to each term of the finite expansion of the symmetric bond averages. Backward incident bonds amount to bounded lattice shifts, and locally uniform convergence on separated bulk configurations permits those shifts. Thus every orientation has the same limiting scalar energy convention. This proves the bounded-domain correlation assertion along the full mesh sequence; the quantity ar_k need not itself converge.

7.5 The periodic thermodynamic limit

The preceding total-background argument does not apply to tori whose side is arbitrarily large compared with r_k . We instead use one additional source without a scaling multiplier. Its contraction controls local expectations even when the total perturbation is large.

Proposition 7.3 (Periodic limits and residual local laws). *After one common decrease of the coupling interval, the following hold.*

1. *At the selected temperature, every bounded observable of finitely many lattice spins has a limit along periodic square boxes as their side tends to infinity. In particular their finite-dimensional laws have a periodic thermodynamic limit at each fixed lattice spacing.*
2. *Let $j \rightarrow \infty$ and $N/r_j \rightarrow \infty$. For each fixed $D < \infty$, the expectations of every test of sup norm at most one supported in a square of radius Dr_j under the residual tilted torus law ν_j^N and the critical nearest-neighbor torus law differ by $o(1)$, uniformly over these tests.*

Proof. We first establish an unscaled marked contraction. Split a short source function into its spin-flip even and odd parts. Combining the response and remainder bounds in Proposition 4.1 bounds the norm of its conditional average modulo constants by $C(CM/L)^{1/8}$ times its sup norm; the odd sector gives the slower decay. Here there is neither a volume multiplier nor the multiplier L^d used for a scaling field. The doubled output weight on its fixed-size carrier costs only a constant. Long source carriers have the exponentially small tail from Proposition 5.2. The total marked norm sums the list directly, so these estimates give

$$\|DR_j(0)u\|_{r_{j+1},\bullet} \leq q_0 \|u\|_{r_j,\bullet}, \quad q_0 < 1, \quad (7.13)$$

after including this condition when choosing the large block ratio. Canonical subtraction of constants costs at most a factor two, already included here. No parity restriction remains on u . The estimate holds on tori through the chart scales in Proposition 6.4, since short cutting squares have the same local critical continuation and long carriers are handled by their norm tail.

By Proposition 5.2, the derivative at a residual background differs by at most $C_L \|H_i^N\|_{r_i}$ from this derivative. The torus bound in Proposition 6.4 and one decrease of $|\lambda|$ therefore give a uniform contraction factor $q_1 < 1$ at every permitted step. Denote the scalar-increment functional for this unscaled derivative by $A_i(H_i^N)$. The same proposition gives

$$\|A_i(H_i^N)\| \leq C_L, \quad \|A_i(H_i^N) - A_i(0)\| \leq C_L \|H_i^N\|_{r_i}. \quad (7.14)$$

Suppose first that a bounded test F is inserted at scale j with support in a square of radius Dr_j . Subtract its uniform-spin mean c and use that square as its marked carrier. The initial marked norm is at most a constant C_D . This carrier may initially be long; the long-carrier part of (7.13) applies to it. Thus D changes the initial bound, not the contraction factor or the coupling interval. Propagate this unscaled derivative through any terminal index s with r_s below the torus chart threshold. Let w_i be its nonscalar list at scale i . The exact first-derivative version of the partition identity reads

$$\mathbb{E}_{\nu_j^N} F = c + \sum_{i=j}^{s-1} A_i(H_i^N) w_i + \mathbb{E}_{\nu_s^N} w_s, \quad \|w_i\|_{r_i,\bullet} \leq C_D q_1^{i-j}. \quad (7.15)$$

The final expectation is bounded by its marked norm, using positivity.

Compare this calculation with the same one on the zero-background trajectory, starting from the same F , and denote its lists by w_i^0 . The operator difference is bounded by $C|\lambda|\gamma^i$. Using a common contraction factor q_1 in both trajectories gives

$$\|w_i - w_i^0\|_{r_i, \bullet} \leq C_D |\lambda| \gamma^j (i - j) q_1^{i-j-1} \quad (i > j).$$

Equations (7.14) and (7.15) can now be summed geometrically. The difference of their scalar sums is at most $C_D |\lambda| \gamma^j$; the two terminal expectations contribute at most $2C_D q_1^{s-j}$. Thus

$$|\mathbb{E}_{\nu_j^N} F - \mathbb{E}_{0,N} F| \leq C_D (|\lambda| \gamma^j + q_1^{s-j}). \quad (7.16)$$

Taking the last permitted s makes $s - j \rightarrow \infty$ whenever $N/r_j \rightarrow \infty$. This proves the second assertion, uniformly over the stated tests.

For the first assertion, fix a bounded observable F of finitely many lattice sites, and choose a fixed connected carrier containing them. Insert it at the microscopic scale. If this carrier does not fit a preliminary cutting square, use the unchanged long-singleton rule, with its protective neighborhood. Its fixed size only changes the finite constants in the preliminary marked bound. Thereafter the unscaled contraction and scalar estimate give a summable tail $C_F \sum_{i \geq m} q_1^i$, uniformly in every sufficiently large torus on which these scales are available.

For any fixed finite set of scales, the background coefficients converge to their plane counterparts by Proposition 6.4. The same is true of the local derivative and each scalar increment. Indeed, lists contained in a common chart agree with the plane list rule; nonchart lists or omitted long inputs have a diameter comparable to N and vanish by the marked diameter estimates. The weighted tails permit the passage from finite lists to the complete coefficient. Thus the first m scalar increments have a limit as $N \rightarrow \infty$. The sum of subsequent increments and the final nonscalar expectation is uniformly negligible as $m \rightarrow \infty$, by the contraction just proved. Applying the telescope from the preliminary scale proves that $\mathbb{E}_{\lambda,N} F$ converges.

Taking F to be cylinder indicators gives consistent limiting finite-dimensional probabilities, hence the asserted thermodynamic state. This construction does not assume uniqueness of other Gibbs states or a choice of a symmetry-broken phase. Its coupling restriction comes from a uniform one-mark operator bound and is independent of F and of the number of sites on which F depends. $\square$

7.6 Completion of the universality theorem

Lemma 7.4 (Mixed reference correlations in the plane). *For every finite configuration of pairwise distinct physical insertion points, use the spin and energy normalizations of (1.2)–(1.3), with each energy centered in its actual nearest-neighbor state. The full-plane critical mixed correlations then have full-sequence scaling limits. These limits are independent of bond directions and equal the limits of their bounded-domain counterparts along an exhaustion by fixed-plus disks, using the actual disk-state means there.*

Proof. Fix separated physical insertion points and a small positive stopping radius below all their pairwise separations. Choose $k = k(a)$ so that ar_k stays between two positive constants below that radius. At zero background, singleton transport is exactly nested conditional averaging. Its nonscalar list is therefore $u_k^\ell = r_k^{d_\ell} \pi_{P_{r_k, z_\ell}} X_\ell$, also for an individual directed bond. By Proposition 7.1 and Lemma 5.3 these lists have bounded marked norm. Hence there is a constant $c_{a,\ell}$ such that

$$f_{a,\ell} = r_k^{d_\ell} P_{r_k, z_\ell} X_\ell - c_{a,\ell}$$

has uniformly bounded sup norm. The constants vanish for spins and are immaterial for centered energies. The cutting squares are disjoint and their union lies in one fixed physical square B_B . These functions are identical in the plane and in every sufficiently large disk D_T containing B_B : each conditional kernel is determined inside its own cutting square.

The conditional Markov identity for every subproduct identifies the original scaled centered correlation with $(ar_k)^{-d(\{1,\dots,q\})}$ times the corresponding centered product of the $f_{a,\ell}$, in either system. Let p_a and $p_{a,T}$ denote these original nearest-neighbor correlations in the plane and in D_T with plus boundary. Conditioning on a common surrounding square and applying Lemma 2.1 with repeated critical RSW in the annuli between B_B and the distant wall gives a total-variation error $\omega(T) \rightarrow 0$ on their common support, uniformly for sufficiently small a . The uniform sup norms control all subset moments. Actual-state centering is a finite polynomial in those moments, so

$$\limsup_{a \downarrow 0} |p_a - p_{a,T}| \leq C\omega(T).$$

For every fixed T , bounded-domain convergence in Proposition 2.5 gives $p_{a,T} \rightarrow \mathcal{C}_{D_T}^+$. Consequently

$$\limsup_{a,b \downarrow 0} |p_a - p_b| \leq 2C\omega(T).$$

Sending T to infinity proves the full mesh limit and identifies it with $\lim_{T \rightarrow \infty} \mathcal{C}_{D_T}^+$. The same comparison proves independence of the bond directions. The order is a fixed positive stopping radius, then the mesh limit at each fixed T , and finally $T \rightarrow \infty$. Only bounded-domain mixed convergence and the preceding uniform conditional-field bounds have been used. $\square$

Proof of Theorem 1.1. Fix the allowed potential once and for all. Proposition 6.2 gives the critical inverse-temperature branch with the specified intercept, positive on a small real coupling interval. The reference perturbation has real coefficients for either sign of the coupling; none of the background estimates requires a sign condition on the finite-range potential. Proposition 7.1 gives the common nonzero normalizations with their unit intercepts. Choose the interval once so that the background, boundary, torus, nonvanishing, and unscaled-contraction bounds used above all hold. These choices do not involve a domain, insertion count, or insertion configuration. Constants in a fixed finite source jet may depend on those data, as the theorem permits.

For any fixed allowed bounded domain, boundary type, and separated configuration, Proposition 7.2 and the bounded-domain evaluation prove the requested full-sequence limit. The microscopic energy sources may have either specified direction. Their plane scalars coincide, and the critical symmetric-average comparison has exactly the common nearest-neighbor energy limit. Equation (7.3) centers every source in its actual finite-domain state. For a fixed negative boundary, the same argument uses its physical prescription throughout; no boundary-dependent normalization has been introduced.

For the full-plane assertion, the first part of Proposition 7.3 establishes the required periodic-box limit at each fixed mesh. Consider an arbitrary sequence $a \rightarrow 0$ and choose its stopping scales by (7.5). On tori with $N/r_k \rightarrow \infty$, Proposition 7.2 still reduces the final source jet to the bounded ideal singleton fields, with vanishing remaining scalar and multiple-source coefficients. Products of any subset of these ideal fields have uniformly bounded sup norms and support in a square of radius Dr_k , for a fixed D depending on the configuration. The second part of Proposition 7.3 therefore replaces their residual tilted expectations by critical torus expectations. This step does not assert smallness of a total background on an expanding torus.

The latter local critical expectations can in turn be replaced by plane expectations as $N/r_k \rightarrow \infty$. Surround their support by a planar chart. The pointwise ratio estimate and repeated critical RSW

in the intervening annuli make the effect of arbitrary spins at the distant chart wall tend to zero, uniformly for the bounded tests. Both the torus and plane laws are mixtures of those continuations. The cutting squares defining the ideal fields are identical local critical squares in both systems. Their simultaneous conditional expectation identity now recovers the original microscopic critical spins and symmetric energies in the plane. Lemma 7.4 finishes the same comparison there. The separately cited plane spin formula identifies its nonzero two-spin amplitude.

To make the order of limits explicit, at each fixed mesh all unscaled moments appearing in the centered product converge as $N \rightarrow \infty$, by Proposition 7.3. The centered moment is a finite polynomial in these moments. Choose N large enough that its error after the fixed mesh normalization is smaller than any prescribed sequence tending to zero, and also so that $N/r_k \rightarrow \infty$. The joint comparison just proved applies to these chosen boxes. Since the original mesh sequence was arbitrary, this proves the stated iterated, full-sequence plane limit. Its two-spin coefficient is nonzero: it is $Z_\sigma(\lambda)^2$ times the positive nearest-neighbor plane amplitude multiplying $|x - y|^{-1/4}$. In particular the selected temperature branch has the massless power-law behavior required here; that critical behavior was a conclusion, not an input to its construction.

Finally let $\phi : D \rightarrow D'$ be as in the theorem. The nearest-neighbor spin field and the energy field centered relative to the plane background have their usual primary covariance, by Proposition 2.5. If $m_D^b(y)$ denotes the domain one-point function of that energy field, it has the same factor $|\phi'(y)|$ as the field itself. Expand a domain-centered mixed correlation as

$$\sum_{T \subseteq \{1, \dots, m\}} (-1)^{m-|T|} \prod_{k \notin T} m_D^b(y_k) \left\langle \prod_{i=1}^n \sigma(x_i) \prod_{k \in T} \epsilon(y_k) \right\rangle_D^b.$$

Every term has the common conformal factor $\prod_i |\phi'(x_i)|^{1/8} \prod_k |\phi'(y_k)|$. Thus the actual-domain-centered nearest-neighbor comparison functions have precisely the covariance stated in the theorem. The constants $Z_\sigma^n Z_\epsilon^m$ do not depend on the domain and preserve this identity. Homogeneous fixed and free boundary types are transported unchanged. This completes every bulk correlation and covariance assertion; no claim concerning coincident or boundary insertions, or interface convergence, is required or used. $\square$

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